

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\401-E-33002-RP.pvdb

Date of Analysis : Aug 12, 2026 9:53am

PV Elite 28, January 2026



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FileName : 401-E-33002-RP

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Vessel Design Summary: Step: 28 9:53am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 600.000 mm ID
Vessel Design Length, Tangent to Tangent 6430.00 mm
Specified Datum Line Distance 76.20 mm
Shell Side Design Temperature 166 °C
Channel Side Design Temperature 66 °C
Shell Side Design Pressure 2600.000 KPa.
Channel Side Design Pressure 2000.000 KPa.
Shop Shell Side Test Pressure 3665.206 KPa.
Shop Channel Side Test Pressure 2637.349 KPa.
Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-106 B	...	...	K03006	
Head	SA-234 WPB	...	...	K03006	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl. cover rea	2005.912	0.00	2029.000	3.1750	No	No
Chnl. rear	2005.912	0.00	2437.000	3.1750	N/A	No
Chnl. flange rer	2000.000	0.00	2272.000	3.1750	N/A	No
Shell	2600.296	0.00	2913.000	3.1750	N/A	No
Chnl. flange frnt	2000.000	0.00	2272.000	3.1750	N/A	No
Chnl. front	2005.912	0.00	2437.000	3.1750	N/A	No
Chnl. front	2005.912	0.00	2029.000	3.1750	No	No

Liquid Level: 600.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
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Vessel Design Summary:

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Ellipse	0.0	76.2	9.5	8.3	8.3	...	1.00	1.00
Cylinder	600.0	600.0	9.5	9.5	8.4	...	1.00	1.00
Body Flg	683.8	79.3	68.0	65.0	61.2	54.8	1.00	1.00
Cylinder	5533.8	4790.0	9.5	9.5	8.8	...	1.00	1.00
Body Flg	5677.6	79.3	68.0	65.0	61.2	54.8	1.00	1.00
Cylinder	6277.6	600.0	9.5	9.5	8.4	...	1.00	1.00
Ellipse	6353.8	76.2	9.5	8.3	8.3	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1- Channel in	WRC-297	0.814	Shell	Passed
N1- Shell outl	WRC-297	0.888	Shell	Passed
N2- shell inle	WRC-297	0.888	Shell	Passed
N2- Channel ou	WRC-297	0.814	Shell	Passed

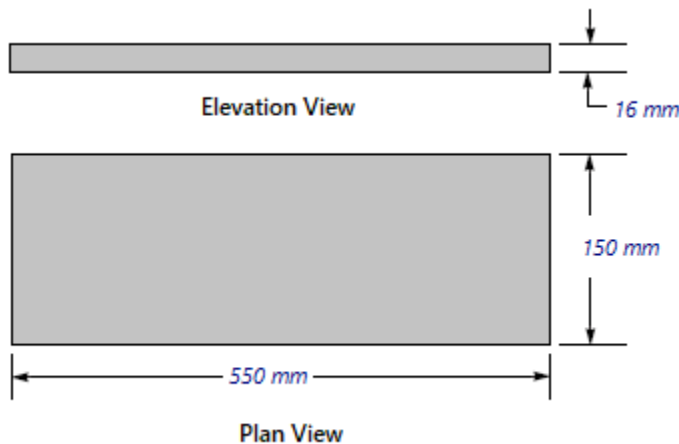
Foundation/Support Summary

Saddle Parameters:

Saddle Width	204.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	560.000 mm
Wear Pad Width	225.000 mm
Wear Pad Thickness	12.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	960.000 mm

Baseplate Length	550.000 mm
Baseplate Thickness	16.000 mm
Baseplate Width	150.000 mm
Number of Ribs (including outside ribs)	3
Rib Thickness	16.000 mm
Web Thickness	16.000 mm
Height of Center Web	306.000 mm
Number of Bolts in Baseplate	2

Baseplate Sketch



Maximum Tensile Bolt Load

10. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:



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Vessel Design Summary: Step: 28 9:53am Aug 12,2026

Maximum Vertical Saddle Load	4132.36	Kgf
Maximum Transverse Saddle Shear Load	532.89	Kgf
Maximum Longitudinal Saddle Shear Load	426.74	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	3477.62	Kgf
Maximum Transverse Saddle Shear Load	175.85	Kgf
Maximum Longitudinal Saddle Shear Load	31.84	Kgf

Weights:

Fabricated - Bare w/o removable internals	4209.4	kg.
Shop Test - Fabricated + water (full)	5800.1	kg.
Shipping - Fab. + rem. intls.+ shipping app.	4276.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	4276.9	kg.
Empty - Fab. + intls. + details + wghts.	4276.9	kg.
Operating - Empty + operating liquid (no ca)	5059.1	kg.

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Nozzle Summary:

Step: 26 9:53am Aug 12,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1- Channel inl	2437	...	...	OK	8.38	OK	Passed
N1- Shell outle	2913	...	...	OK	8.84	OK	Passed
N2- shell inlet	2819	...	...	OK	8.84	OK	Passed
N2- Channel out	2437	...	...	OK	8.37	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2819 Nozzle : N2- shell inlet [Shellside]
 Minimum MAWP Nozzles : 2437 Nozzle : N1- Channel inl [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1- Channel inle	376.200	90.000	305.359
40	N1- Shell outlet	1320.000	270.000	498.450
40	N2- shell inlet	4820.000	270.000	590.601
60	N2- Channel outl	6053.799	270.000	305.359

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 25 9:53am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1- Channel inle	6.000 in	80	WNF	6.625	10.973	308.27	10.00	271.57	300
N2- Channel outl	6.000 in	80	WNF	6.625	10.973	308.27	10.00	271.57	300
N1- Shell outlet	10.000 in	80	WNF	10.750	15.088	413.05	10.00	292.40	300
N2- shell inlet	12.000 in	80	WNF	12.750	17.450	463.85	10.00	306.98	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1- Channel	SA-106 B	9.530	30.000	8.000	10.000	...
N2- Channel	SA-106 B	9.530	30.000	8.000	10.000	...
N1- Shell o	SA-106 B	9.530	10.000	8.000	10.000	...
N2- shell i	SA-106 B	9.530	10.000	8.000	10.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1- Channel inle	300.000	90.0	250.00	0.00	Chnl. rear
N2- Channel outl	5977.599	270.0	250.00	0.00	Chnl. front
N1- Shell outlet	1243.800	270.0	250.00	0.00	Shell
N2- shell inlet	4743.799	270.0	250.00	0.00	Shell

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Bill of Material:

Step: 24 9:53am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 9.5mm THK X 600.0mm ID X 76.2mm	SA-234 WPB
2	WELD NECK FLANGE: 68.0mm THK X 740.0mm OD	SA-266 2
8	BAFFLES: 580.0mm DIA X 6.0mm THK	SA-516 70
2	TUBESHEET: 740.0mm X 60.0mm THK	SA-266 2
583	TUBES: 4877.0mm X 15.9mm DIA X 2.1mm THK	SA-179
2	SADDLE: 204.0mm X 120 DEG	SA-516 70
1	INSULATION: 4790mm X 45mm THK	Hot
2	GASKET: 655.0mm OD X 629.0mm ID	...
56	BODY FLANGE BOLTS: 19.0mm DIA	SA-193 B7
56	NUTS FOR BODY FLANGE BOLTS: 19.0mm DIA	...
2	CYLINDRICAL SEGMENT 9.5mm THK X 748.7mm X 600.0mm	SA-106 B
1	CYLINDRICAL SEGMENT 9.5mm THK X 387.5mm X 600.0mm	SA-106 B
1	CYLINDRICAL SEGMENT 9.5mm THK X 1513.2mm X 4790.0mm	SA-106 B
1	CYLINDRICAL SEGMENT 9.5mm THK X 371.7mm X 4790.0mm	SA-106 B
1	CYLINDRICAL SEGMENT 9.5mm THK X 371.7mm X 600.0mm	SA-106 B
1	CYLINDRICAL SEGMENT 9.5mm THK X 1513.2mm X 600.0mm	SA-106 B
1	NAMEPLATE	...

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MDMT Summary:

Step: 27 9:53am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell	[8]	B	-29	-48	-29	0.416	9.530	1.00	No
Left Saddle	[24]	B	-29	-38	-29	0.844	9.530	1.00	No
Right saddle	[24]	B	-29	-38	-29	0.844	9.530	1.00	No
N1- Shell outl	[1]	B	-29	-41	-29	0.778	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N2- shell inle	[1]	B	-29	-41	-29	0.778	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: SS	[13]	B	-16	-24	-29	0.856	15.000	1.00	No

Warmest MDMT: -16 -24

Chnl. flange	[11]	B	-29	-38	-29	0.831	9.530	1.00	No
Chnl. flange	[11]	B	-29	-38	-29	0.831	9.530	1.00	No
Chnl. cover r	[10]	B	-29	-29	-29	0.997	8.340	1.00	No
Chnl. cover re	[7]	B	-29	-38	-29	0.829	9.530	1.00	No
Chnl. rear	[8]	B	-29	-48	-29	0.503	8.340	1.00	No
Chnl. front	[8]	B	-29	-48	-29	0.409	9.530	1.00	No
Chnl. front	[10]	B	-29	-29	-29	0.997	8.340	1.00	No
Chnl. front	[7]	B	-29	-38	-29	0.829	9.530	1.00	No
N1- Channel in	[1]	B	-29	-39	-29	0.819	9.530	1.00	No
Nozzle Flg	[5]	A	-18	-48					
N2- Channel ou	[1]	B	-29	-39	-29	0.817	9.530	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tubesheet: CS	[14]	B	-16	-24	-29	0.856	15.000	1.00	No
Bolting	[21]		-48						
Bolting	[21]		-48						

Warmest MDMT: -16 -24

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	11.4	Pass
Channel/Tube	-29.0	11.4	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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MDMT Summary:

Step: 27 9:53am Aug 12, 2026

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 9:53am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 9:53am Aug 12,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2600	KPa.
Channel Side Design Pressure	2000	KPa.
Shell Side Design Temperature	165.6	°C
Channel Side Design Temperature	66.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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Input Echo:

Step: 1 9:53am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test

UG-99(b)

Pressure Test Position

Horizontal

Test Performed in Corroded Condition

No

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000



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Input Echo: Step: 1 9:53am Aug 12,2026

Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/7
Element From Node	10
Element To Node	20
Element Type	Elliptical



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Input Echo:

Step: 1 9:53am Aug 12,2026

Description	Chnl. cover rea	
Distance "FROM" to "TO"	76.2	mm
Inside Diameter	600	mm
Element Thickness	8.34	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	9.53	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2000	KPa.
Design Temperature Internal Pressure	66	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-234 WPB	
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	153.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	226.8	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. & wld. fittings	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	600	mm
Liquid Density	0.0009984	kg./cm ³

Element#	2/7
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Chnl. rear
Distance "FROM" to "TO"	600 mm
Inside Diameter	600 mm
Element Thickness	9.53 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	9.53 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2000 KPa.
Design Temperature Internal Pressure	66 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-106 B
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	153.4 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	226.8 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. pipe
Efficiency, Longitudinal Seam	1.0



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Input Echo:

Step: 1 9:53am Aug 12,2026

Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	600 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	20
Detail Type	Nozzle
Detail ID	N1- Channel inle
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	6 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	90.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	27.062 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	3/7
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Chnl. flange rer
Distance "FROM" to "TO"	79.3 mm
Flange Inside Diameter	600 mm
Element Thickness	65 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2000 KPa.
Design Temperature Internal Pressure	66 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	179.4 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	232.83 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	4/7
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	4790 mm
Inside Diameter	600 mm
Element Thickness	9.53 mm



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Input Echo:

Step: 1 9:53am Aug 12,2026

Internal Corrosion Allowance	3.175	mm
Nominal Thickness	9.53	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2600	KPa.
Design Temperature Internal Pressure	166	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-106 B	
Allowable Stress, Ambient	161	N./mm ²
Allowable Stress, Operating	140.75	N./mm ²
Allowable Stress, Hydrotest	228	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	0	mm
Yield Stress, Operating	211.51	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. pipe	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	960	mm
Width of Saddle	204	mm
Height of Saddle at Bottom	560	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	225	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	3830	mm
Width of Saddle	204	mm
Height of Saddle at Bottom	560	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	225	mm
Thickness of Wear Plate	12	mm
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.5	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	600	mm
Liquid Density	0.49920E-04	kg./cm ³



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Input Echo:

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Element From Node	40	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	4790	mm
Thickness of Insulation	45	mm
Density	0.00015	kg./cm ³

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N1- Shell outlet	
Dist. from "FROM" Node / Offset dist	500	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	78.846	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N2- shell inlet	
Dist. from "FROM" Node / Offset dist	4000	mm
Nozzle Diameter	12	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	110.13	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	5/7	
Element From Node	50	
Element To Node	60	
Element Type	Flange	
Description	Chnl. flange frnt	
Distance "FROM" to "TO"	79.3	mm
Flange Inside Diameter	600	mm
Element Thickness	65	mm
Internal Corrosion Allowance	3.175	mm
Nominal Thickness	68	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2000	KPa.
Design Temperature Internal Pressure	66	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	179.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	232.83	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	



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Input Echo:

Step: 1 9:53am Aug 12,2026

Weld is pre-Heated

No

Element#	6/7
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Chnl. front
Distance "FROM" to "TO"	600 mm
Inside Diameter	600 mm
Element Thickness	9.53 mm
Internal Corrosion Allowance	3.175 mm
Nominal Thickness	9.53 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2000 KPa.
Design Temperature Internal Pressure	66 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-106 B
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	153.4 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	226.8 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. pipe
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	600 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	N2- Channel outl
Dist. from "FROM" Node / Offset dist	300 mm
Nozzle Diameter	6 in.
Nozzle Schedule	80
Nozzle Class	300
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	27.062 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	7/7
Element From Node	70
Element To Node	80
Element Type	Elliptical
Description	Chnl. front
Distance "FROM" to "TO"	76.2 mm
Inside Diameter	600 mm
Element Thickness	8.34 mm



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Input Echo:

Step: 1 9:53am Aug 12,2026

Internal Corrosion Allowance	3.175	mm
Nominal Thickness	9.53	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2000	KPa.
Design Temperature Internal Pressure	66	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-234	WPB
Allowable Stress, Ambient	118	N./mm ²
Allowable Stress, Operating	118	N./mm ²
Allowable Stress, Hydrotest	153.4	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	226.8	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03006	
Product Form	Smls. & wld. fittings	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 70	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	600	mm
Liquid Density	0.0009984	kg./cm ³

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XY Coordinate Calculations: Step: 2 9:53am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Chnl. cover rea		76.2	...	76.2	...
Chnl. rear		676.2	...	600	...
Chnl. flange re		760	...	79.3	...
Shell		5610	...	4790	...
Chnl. flange fr		5753.8	...	79.3	...
Chnl. front		6353.8	...	600	...
Chnl. front		6430	...	76.2	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: FLANGE

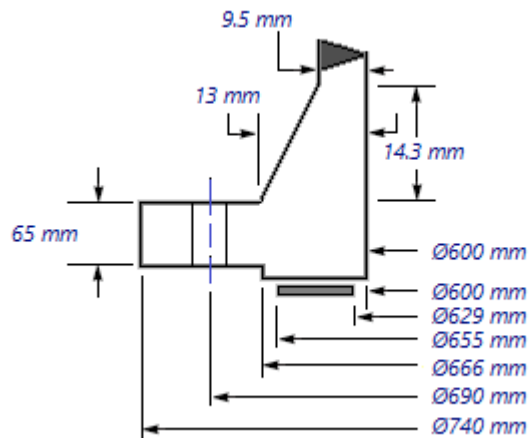
Flng: 5 9:53am Aug 12,2026

Flange Input Data Values

Description: FLANGE :

Chnl. flange rer

Flange Type	Integral Weld Neck		
Design Pressure	P	2000.00	KPa.
Design Temperature		66	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	600.000	mm
Flange Outside Diameter	A	740.000	mm
Flange Thickness	t	65.0000	mm
Thickness of Hub at Small End	go	9.5300	mm
Thickness of Hub at Large End	gl	13.0000	mm
Length of Hub	h	14.3000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	690.000	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		28	
Flange Face Outside Diameter	Fod	666.000	mm
Flange Face Inside Diameter	Fid	600.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	655.000	mm
Gasket Inside Diameter	Gi	629.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: FLANGE

Flng: 5 9:53am Aug 12, 2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2000.0 \cdot (600.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 2000.0) + Ca$$

$$= 7.6077 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355)$$

$$= 2856.539 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.53) / (300.0 + 0.6 \cdot 9.53)$$

$$= 4301.480 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	61.825 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	606.350 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	9.825 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	6.355 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	32.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	642.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 642.1509^2 \cdot 2000.0$$

$$= 66052.695 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 642.1509 \cdot 3.0 \cdot 2000.0$$

$$= 15860.175 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 606.35^2 \cdot 2000.0$$

$$= 58892.934 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 66053 - 58893$$

$$= 7159.764 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(66053 + 15860 + 0, 0)$$

$$= 81912.875 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 642.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 91124.633 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((81913 + 0 + 4 \cdot 0/642.15)/172, 91125/172)$$

$$= 51.957 \text{ cm}^2$$



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Flg Calc [Int P]: FLANGE

Flng: 5 9:53am Aug 12,2026

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 19.05 + 6 \cdot 61.825/(3.0 + 0.5)$$

$$= 144.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 690.0 \cdot \sin(\pi/28)$$

$$= 77.255 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{77.255/(2 \cdot 19.05 + 61.825)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	51.957	54.555	
Radial Distance between Hub and Bolts:	28.575	32.000	
Radial Distance between Bolts and Edge:	20.637	25.000	
Circ. Spacing between the Bolts:	44.450	77.255	144.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 54.555 \cdot 172.0 / (68.95 \cdot \pi(655.0 + 629.0))$$

$$= 3.374 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(51.957 + 54.5547) / 2$$

$$= 93402.66 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 81913 - 66053$$

$$= 15860.17 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (690.0 - 642.1509) / 2$$

$$= 23.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((690.0 - 606.35) / 2 + 23.9246)$$

$$= 32.8748 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (690.0 - 606.35 - 9.825) / 2$$

$$= 36.9125 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	58893.	36.9125	1.0000	2174.
Face Pressure, Mt	7160.	32.8748	1.0000	235.
Gasket Load, Mg	15860.	23.9246	1.0000	379.
Gasket Seating, Matm	93403.	23.9246	1.0000	2235.



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Flg Calc [Int P]: FLANGE

Flng: 5 9:53am Aug 12, 2026

Total Moment for Operation, Mo 2789. Kg-m.
Total Moment for Gasket seating, Mg 2235. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 62.075 mm
Hub Ratio, h/h0 = HL / H0 0.230
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.546

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.894
Factor V per 4.16.5 0.391
Factor f per 4.16.5 1.483

Stress factor from Table 4.16.4 [K]:

= A / B
= 740.0/606.35
= 1.220

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1))
= (0.66845 + (5.7169(1.22² • log10(1.22)) / (1.22² - 1)) / (1.22 - 1))
= 9.861

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.22²(1 + 8.55246 • log10(1.22)) - 1) / ((1.0472 + 1.9448 • 1.22²) • (1.22 - 1))
= 1.831

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.22²(1 + 8.55246 • log10(1.22)) - 1) / (1.36136(1.22² - 1) • (1.22 - 1))
= 10.836

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.22² + 1) / (1.22² - 1)
= 5.086

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 10.836 • 6.355² • 62.075 / 0.391
= 69423.953 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.894/62.075
= 0.014 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (61.825 • 0.014 + 1) / 1.831 + 61.825³/69423.953
= 4.437

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.4827•2789/606.35)/(4.4365•9.825²)
= 156.16 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1.4827•2235/606.35)/(4.4365•9.825²)
= 125.13 N./mm²



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Flg Calc [Int P]: FLANGE

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Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.187 \cdot 2789 / 606.35) / (4.4365 \cdot 61.825^2)$$

$$= 5.82 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.187 \cdot 2235 / 606.35) / (4.4365 \cdot 61.825^2)$$

$$= 4.66 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.8609 \cdot 2789 / (61.825^2 \cdot 606.35)) - 5.0865 \cdot 6$$

$$= 86.77 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.8609 \cdot 2235 / (61.825^2 \cdot 606.35)) - 5.0865 \cdot 5$$

$$= 69.53 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (156 + \max(6, 87)) / 2$$

$$= 121.46 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (125 + \max(5, 70)) / 2$$

$$= 97.33 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 81913 / 54.5547$$

$$= 147.25 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (91125 / 54.5547)$$

$$= 163.81 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	156.16	207.00	125.13	207.00
Radial Flange	5.82	138.00	4.66	138.00
Tangential Flange	86.77	138.00	69.53	138.00
Maximum Average	121.46	138.00	97.33	138.00
Bolting	147.25	172.00	163.81	172.00

Minimum Required Flange Thickness 61.189 mm

Estimated M.A.W.P. (Operating) 2272 KPa.

Estimated Finished Weight of Flange at given Thk. 76.6 kg.

Estimated Unfinished Weight of Forging at given Thk 90.6 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2234.7 / 1.0 \cdot 9806.637 \cdot 0.391 / (3.815 \cdot 202713 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.768 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:



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Flg Calc [Int P]: FLANGE

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$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2788.8 / 1.0 \cdot 9806.637 \cdot 0.391 / (3.815 \cdot 200142 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.971 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2234.7 / 1.0 \cdot 9806.637 \cdot 0.391 / (4.437 \cdot 202713 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.661 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2788.8 / 1.0 \cdot 9806.637 \cdot 0.391 / (4.437 \cdot 200142 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.835 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.831, Temperature Reduction per Fig. UCS 66.1 = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	133.36	207.00	103.17	207.00
Radial Flange	8.49	138.00	6.57	138.00
Tangential Flange	63.75	138.00	49.31	138.00
Maximum Average	98.55	138.00	76.24	138.00
Bolting	147.25	172.00	163.81	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 52.426 mm

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Flg Calc [Int P]: New Flange

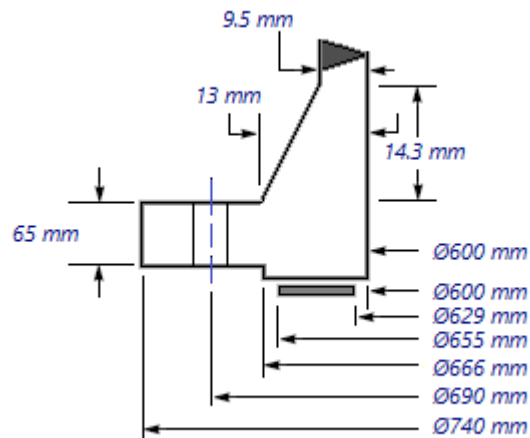
Flng: 6 9:53am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl. flange frnt

Flange Type	Integral Weld Neck		
Design Pressure	P	2000.00	KPa.
Design Temperature		66	°C
Internal Corrosion Allowance	ci	3.1750	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	600.000	mm
Flange Outside Diameter	A	740.000	mm
Flange Thickness	t	65.0000	mm
Thickness of Hub at Small End	go	9.5300	mm
Thickness of Hub at Large End	gl	13.0000	mm
Length of Hub	h	14.3000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	690.000	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		28	
Flange Face Outside Diameter	Fod	666.000	mm
Flange Face Inside Diameter	Fid	600.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	655.000	mm
Gasket Inside Diameter	Gi	629.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm
Raised face depth	Fd	0.0000	mm





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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2000.0 \cdot (600.0/2 + 3.175)) / (138.0 \cdot 1.0 - 0.6 \cdot 2000.0) + Ca$$

$$= 7.6077 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355)$$

$$= 2856.539 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.53) / (300.0 + 0.6 \cdot 9.53)$$

$$= 4301.480 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	61.825 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	606.350 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	9.825 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	6.355 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	32.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm
Basic Gasket Width, $bo = N/2$	6.500 mm
Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$	6.425 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	642.151 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 642.1509^2 \cdot 2000.0$$

$$= 66052.695 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 642.1509 \cdot 3.0 \cdot 2000.0$$

$$= 15860.175 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 606.35^2 \cdot 2000.0$$

$$= 58892.934 \text{ Kgf}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 66053 - 58893$$

$$= 7159.764 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(66053 + 15860 + 0, 0)$$

$$= 81912.875 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 642.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 91124.633 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((81913 + 0 + 4 \cdot 0/642.15)/172, 91125/172)$$

$$= 51.957 \text{ cm}^2$$



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Flg Calc [Int P]: New Flange

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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 19.05 + 6 \cdot 61.825/(3.0 + 0.5)$$

$$= 144.086 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 690.0 \cdot \sin(\pi/28)$$

$$= 77.255 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{Bs/(2a + t)}, 1)$$

$$= \max(\sqrt{77.255/(2 \cdot 19.05 + 61.825)}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	51.957	54.555	
Radial Distance between Hub and Bolts:	28.575	32.000	
Radial Distance between Bolts and Edge:	20.637	25.000	
Circ. Spacing between the Bolts:	44.450	77.255	144.086

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 54.555 \cdot 172.0 / (68.95 \cdot \pi(655.0 + 629.0))$$

$$= 3.374 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(51.957 + 54.5547) / 2$$

$$= 93402.66 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 81913 - 66053$$

$$= 15860.17 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (690.0 - 642.1509) / 2$$

$$= 23.9246 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((690.0 - 606.35) / 2 + 23.9246)$$

$$= 32.8748 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (690.0 - 606.35 - 9.825) / 2$$

$$= 36.9125 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	58893.	36.9125	1.0000	2174.
Face Pressure, Mt	7160.	32.8748	1.0000	235.
Gasket Load, Mg	15860.	23.9246	1.0000	379.
Gasket Seating, Matm	93403.	23.9246	1.0000	2235.



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Flg Calc [Int P]: New Flange

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Total Moment for Operation, Mo 2789. Kg-m.
Total Moment for Gasket seating, Mg 2235. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 62.075 mm
Hub Ratio, h/h0 = HL / H0 0.230
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.546

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.894
Factor V per 4.16.5 0.391
Factor f per 4.16.5 1.483

Stress factor from Table 4.16.4 [K]:

= A / B
= 740.0/606.35
= 1.220

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169(1.22² • log10(1.22)) / (1.22² - 1)) / (1.22 - 1)
= 9.861

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
= (1.22²(1 + 8.55246 • log10(1.22)) - 1) / ((1.0472 + 1.9448 • 1.22²) • (1.22 - 1))
= 1.831

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
= (1.22²(1 + 8.55246 • log10(1.22)) - 1) / (1.36136(1.22² - 1) • (1.22 - 1))
= 10.836

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.22² + 1) / (1.22² - 1)
= 5.086

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
= 10.836 • 6.355² • 62.075 / 0.391
= 69423.953 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.894/62.075
= 0.014 1/mm

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
= (61.825 • 0.014 + 1) / 1.831 + 61.825³/69423.953
= 4.437

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
= (1.4827•2789/606.35)/(4.4365•9.825²)
= 156.16 N./mm²

Longitudinal Hub Stress, Seating [SHa]:

= (f • Matm / Bcor) / (L • g1²)
= (1.4827•2235/606.35)/(4.4365•9.825²)
= 125.13 N./mm²



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Flg Calc [Int P]: New Flange

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Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.187 \cdot 2789 / 606.35) / (4.4365 \cdot 61.825^2)$$

$$= 5.82 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.187 \cdot 2235 / 606.35) / (4.4365 \cdot 61.825^2)$$

$$= 4.66 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (9.8609 \cdot 2789 / (61.825^2 \cdot 606.35)) - 5.0865 \cdot 6$$

$$= 86.77 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (9.8609 \cdot 2235 / (61.825^2 \cdot 606.35)) - 5.0865 \cdot 5$$

$$= 69.53 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (156 + \max(6, 87)) / 2$$

$$= 121.46 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (125 + \max(5, 70)) / 2$$

$$= 97.33 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 81913 / 54.5547$$

$$= 147.25 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (91125 / 54.5547)$$

$$= 163.81 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	156.16	207.00	125.13	207.00
Radial Flange	5.82	138.00	4.66	138.00
Tangential Flange	86.77	138.00	69.53	138.00
Maximum Average	121.46	138.00	97.33	138.00
Bolting	147.25	172.00	163.81	172.00

Minimum Required Flange Thickness 61.189 mm

Estimated M.A.W.P. (Operating) 2272 KPa.

Estimated Finished Weight of Flange at given Thk. 76.6 kg.

Estimated Unfinished Weight of Forging at given Thk 90.6 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2234.7 / 1.0 \cdot 9806.637 \cdot 0.391 / (3.815 \cdot 202713 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.768 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:



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Flg Calc [Int P]: New Flange Flng: 6 9:53am Aug 12,2026

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2788.8 / 1.0 \cdot 9806.637 \cdot 0.391 / (3.815 \cdot 200142 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.971 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2234.7 / 1.0 \cdot 9806.637 \cdot 0.391 / (4.437 \cdot 202713 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.661 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 2788.8 / 1.0 \cdot 9806.637 \cdot 0.391 / (4.437 \cdot 200142 \cdot 6.355^2 \cdot 62.075 \cdot 0.3)$$

$$= 0.835 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.831, Temperature Reduction per Fig. UCS 66.1 = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	133.36	207.00	103.17	207.00
Radial Flange	8.49	138.00	6.57	138.00
Tangential Flange	63.75	138.00	49.31	138.00
Maximum Average	98.55	138.00	76.24	138.00
Bolting	147.25	172.00	163.81	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 52.426 mm

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 Internal Pressure Calculations: Step: 5 9:53am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Chnl. cover rea		2005.9	9.53	3.175	600	118	1
Chnl. rear		2005.9	9.53	3.175	600	118	2
Chnl. flange rer		2000	68	3.175	600	138	3
Shell		2600.3	9.53	3.175	600	140.75	4
Chnl. flange frn		2000	68	3.175	600	138	5
Chnl. front		2005.9	9.53	3.175	600	118	6
Chnl. front		2005.9	9.53	3.175	600	118	7

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Chnl. cover rea		2000	2028.73	3271.06	8.34	8.26694	1
Chnl. rear		2000	2436.64	3678.08	9.53	8.38226	2
Chnl. flange rer		2000	2272.22	2336.16	65	61.1886	3
Shell		2600	2913.3	5018.39	9.53	8.83899	4
Chnl. flange frn		2000	2272.22	2336.16	65	61.1886	5
Chnl. front		2000	2436.64	3678.08	9.53	8.38226	6
Chnl. front		2000	2028.73	3271.06	8.34	8.26694	7

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 2819 KPa.
 Shell Side MAPnc = 5018 KPa.
 Channel Side MAWP = 2029 KPa.
 Channel Side MAPnc = 2336 KPa.

Elements Suitable for Design Internal Pressure.

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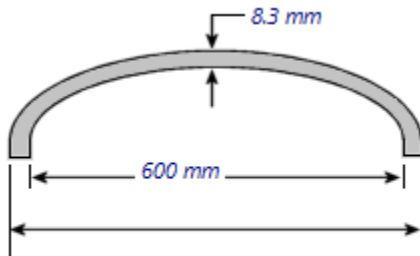
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Element: Chnl. cover rea

Step: 6 9:53am Aug 12,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Chnl. cover rea

Elliptical Head Design Information:			
Design Pressure P	2005.912 KPa.	Int. Design Temperature	66 °C
Material	SA-234 WPB	Ext. Design Temperature	40 °C
Length	76.200 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.1750 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	600.000 mm	Actual Stress	116.3 N./mm ²
Spec. Min. Thk t,e, or sa	8.3400 mm	Spec. Nominal Thk.	9.530 mm
Surface Area	5636.2 cm ²	SG of Contents	1.000
Element Volume	49828.3 cm ³	Weight of Contents	49.8 Kg
Empty Weight	43.6 Kg	Operating Weight	93.4 Kg
		Comp. All. Ope. Stress	125.7 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2005.912 \cdot 606.35 \cdot 0.986) / (2 \cdot 118.0 \cdot 1.0 - 0.2 \cdot 2005.912)$$

$$= 5.0919 + 3.1750 = 8.2669 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.912 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 5.165) / (0.986 \cdot 606.35 + 0.2 \cdot 5.165)$$

$$= 2035 - 6 = 2029 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 8.34) / (1.0 \cdot 600.0 + 0.2 \cdot 8.34)$$

$$= 3271 \text{ KPa.}$$



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Element: Chnl. cover rea

Step: 6 9:53am Aug 12,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2005.912 \cdot (0.986 \cdot 606.35 + 0.2 \cdot 5.165)) / (2 \cdot 1.0 \cdot 5.165)$$

$$= 116.332 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2005.912 \cdot 303.175) / (118.0 \cdot 1.0 - 0.6 \cdot 2005.912) + 3.175$$

$$= 8.382 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 5.912 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355)$$

$$= 2443 - 6 = 2437 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (606.35 / (2 \cdot 153.175))^2) / 6$$

$$= 0.986253$$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8.34, tr = 5.15, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.997, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 9.53, tr = 5.267, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.829, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2028.730 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	9.530 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.829	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

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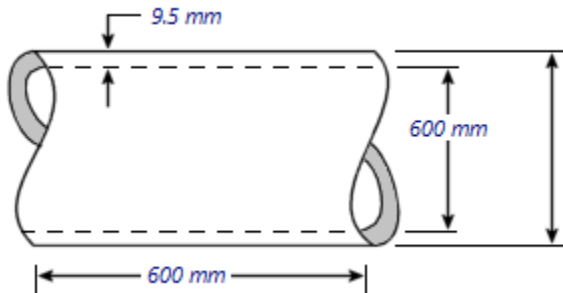
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Element: Chnl. rear

Step: 7 9:53am Aug 12,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:			
Design Pressure P	2005.912 KPa.	Int. Design Temperature	66 °C
Material	SA-106 B	Ext. Design Temperature	40 °C
Length	600.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.1750 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	600.000 mm	Actual Stress	96.9 N./mm ²
Spec. Min. Thk t _e , or sa	9.5300 mm	Spec. Nominal Thk.	9.530 mm
Surface Area	11669.0 cm ²	SG of Contents	1.000
Element Volume	169676.4 cm ³	Weight of Contents	169.6 Kg
Empty Weight	84.9 Kg	Operating Weight	254.4 Kg
		Comp. All. Ope. Stress	130.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2005.912 \cdot 303.175) / (118.0 \cdot 1.0 - 0.6 \cdot 2005.912) \\
 &= 5.2073 + 3.1750 = 8.3823 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$\begin{aligned}
 &\text{Less Operating Hydrostatic Head Pressure of 5.912 KPa.} \\
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118.0 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355) \\
 &= 2443 - 6 = 2437 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (118.0 \cdot 1.0 \cdot 9.53) / (300.0 + 0.6 \cdot 9.53) \\
 &= 3678 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2005.912 \cdot (303.175 + 0.6 \cdot 6.355)) / (1.0 \cdot 6.355) \\
 &= 96.904 \text{ N./mm}^2
 \end{aligned}$$



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Element: Chnl. rear

Step: 7 9:53am Aug 12,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 8.34, tr = 2.597, c = 3.175 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.503$, Temp. Reduction = 32 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2028.730 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	8.340 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.503	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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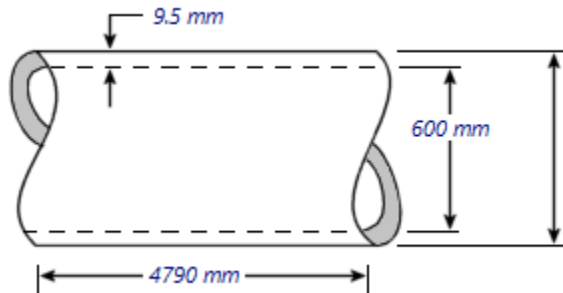
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Element: Shell

Step: 8 9:53am Aug 12,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	2600.296 KPa.	Int. Design Temperature	166 °C
Material	SA-106 B	Ext. Design Temperature	40 °C
Length	4790.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.1750 mm	Allowable Stress (ope) S	140.8 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	161.0 N./mm ²
Inside Diameter Di	600.000 mm	Actual Stress	125.6 N./mm ²
Spec. Min. Thk t _e , or sa	9.5300 mm	Spec. Nominal Thk.	9.530 mm
Surface Area	93157.6 cm ²	SG of Contents	0.050
Element Volume	801743.4 cm ³	Weight of Contents	40.1 Kg
Empty Weight	677.5 Kg	Operating Weight	717.6 Kg
		Comp. All. Ope. Stress	126.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2600.296 \cdot 303.175) / (140.75 \cdot 1.0 - 0.6 \cdot 2600.296)$$

$$= 5.6640 + 3.1750 = 8.8390 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.296 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (140.75 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355)$$

$$= 2914 - 0 = 2913 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (161.0 \cdot 1.0 \cdot 9.53) / (300.0 + 0.6 \cdot 9.53)$$

$$= 5018 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2600.296 \cdot (303.175 + 0.6 \cdot 6.355)) / (1.0 \cdot 6.355)$$

$$= 125.619 \text{ N./mm}^2$$



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Element: Shell

Step: 8 9:53am Aug 12,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 9.53, tr = 2.645, c = 3.175 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.416$, Temp. Reduction = 47 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

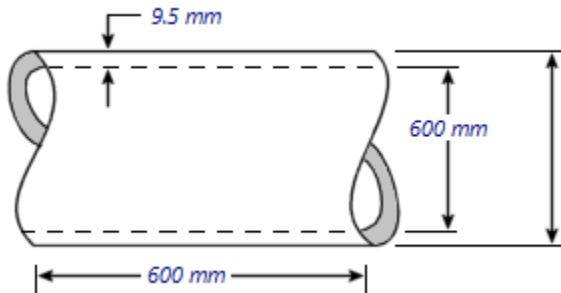
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2819.390 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	9.530 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.416	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Chnl. front

Cylindrical Shell Design Information:			
Design Pressure P	2005.912 KPa.	Int. Design Temperature	66 °C
Material	SA-106 B	Ext. Design Temperature	40 °C
Length	600.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.1750 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	600.000 mm	Actual Stress	96.9 N./mm ²
Spec. Min. Thk t,e, or sa	9.5300 mm	Spec. Nominal Thk.	9.530 mm
Surface Area	11669.0 cm ²	SG of Contents	1.000
Element Volume	169676.4 cm ³	Weight of Contents	169.6 Kg
Empty Weight	84.9 Kg	Operating Weight	254.4 Kg
		Comp. All. Ope. Stress	130.8 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2005.912 \cdot 303.175) / (118.0 \cdot 1.0 - 0.6 \cdot 2005.912)$$

$$= 5.2073 + 3.1750 = 8.3823 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.912 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355)$$

$$= 2443 - 6 = 2437 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 9.53) / (300.0 + 0.6 \cdot 9.53)$$

$$= 3678 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (2005.912 \cdot (303.175 + 0.6 \cdot 6.355)) / (1.0 \cdot 6.355)$$

$$= 96.904 \text{ N./mm}^2$$



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Element: Chnl. front

Step: 9 9:53am Aug 12,2026

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 9.53, tr = 2.597, c = 3.175 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.409$, Temp. Reduction = 48 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2028.730 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	9.530 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.409	Temperature Difference	19 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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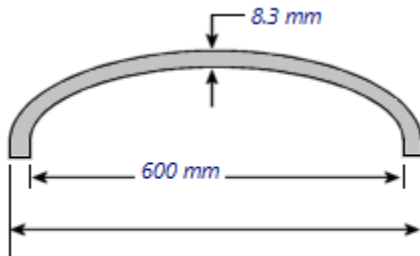
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Element: Chnl. front

Step: 10 9:53am Aug 12,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Chnl. front

Elliptical Head Design Information:			
Design Pressure P	2005.912 KPa.	Int. Design Temperature	66 °C
Material	SA-234 WPB	Ext. Design Temperature	40 °C
Length	76.200 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.1750 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	600.000 mm	Actual Stress	116.3 N./mm ²
Spec. Min. Thk t,e, or sa	8.3400 mm	Spec. Nominal Thk.	9.530 mm
Surface Area	5636.2 cm ²	SG of Contents	1.000
Element Volume	49828.3 cm ³	Weight of Contents	49.8 Kg
Empty Weight	43.6 Kg	Operating Weight	93.4 Kg
		Comp. All. Ope. Stress	130.8 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2005.912 \cdot 606.35 \cdot 0.986) / (2 \cdot 118.0 \cdot 1.0 - 0.2 \cdot 2005.912)$$

$$= 5.0919 + 3.1750 = 8.2669 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 5.912 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 5.165) / (0.986 \cdot 606.35 + 0.2 \cdot 5.165)$$

$$= 2035 - 6 = 2029 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 8.34) / (1.0 \cdot 600.0 + 0.2 \cdot 8.34)$$

$$= 3271 \text{ KPa.}$$



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Element: Chnl. front

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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2005.912 \cdot (0.986 \cdot 606.35 + 0.2 \cdot 5.165)) / (2 \cdot 1.0 \cdot 5.165)$$

$$= 116.332 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2005.912 \cdot 303.175) / (118.0 \cdot 1.0 - 0.6 \cdot 2005.912) + 3.175$$

$$= 8.382 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 5.912 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 6.355) / (303.175 + 0.6 \cdot 6.355)$$

$$= 2443 - 6 = 2437 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (606.35 / (2 \cdot 153.175))^2) / 6$$

$$= 0.986253$$

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 8.34, tr = 5.15, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.997, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 9.53, tr = 5.267, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.829, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2028.730 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	9.530 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.829	Temperature Difference	10 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.500 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.750 mm

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Shop Pressure Test Results: Step: 11 9:53am Aug 12,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3665.206 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3380.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	6518.030 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	3101.329 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3718.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3718.000 KPa.
App 27-4	= M.A.W.P.	2819.390 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2637.349 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	2600.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3031.124 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2231.603 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	2860.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	2860.000 KPa.
App 27-4	= M.A.W.P.	2028.730 KPa.

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Chnl. cover rea	95.4	153.4	0.622	2643.23
Chnl. rear	84.8	153.4	0.553	2643.23
Shell	117.8	228.0	0.517	3671.09
Chnl. front	84.8	153.4	0.553	2643.23
Chnl. front	95.4	153.4	0.622	2643.23

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1- Channel inle	Nozzle	118.00	118.00	1.000
N1- Channel inle	Pad	138.00	138.00	1.000
N1- Shell outlet	Nozzle	118.00	118.00	1.000
N1- Shell outlet	Pad	138.00	138.00	1.000
N2- shell inlet	Nozzle	118.00	118.00	1.000
N2- shell inlet	Pad	138.00	138.00	1.000
N2- Channel outl	Nozzle	118.00	118.00	1.000
N2- Channel outl	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Chnl. cover rea	118.00	118.00	1.000
Chnl. rear	118.00	118.00	1.000
Chnl. flange rer	138.00	138.00	1.000
Shell	161.00	140.75	1.144
Chnl. flange frnt	138.00	138.00	1.000
Chnl. front	118.00	118.00	1.000



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Shop Pressure Test Results: Step: 11 9:53am Aug 12,2026

Chnl. front	118.00	118.00	1.000
-------------	--------	--------	-------

Minimum	1.000
---------	-------

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum	1.000
---------	-------

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1- Channel inle	19.21	153.40	0.125
N1- Shell outlet	31.75	153.40	0.207
N2- shell inlet	32.60	153.40	0.213
N2- Channel outl	19.21	153.40	0.125

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 9:53a Aug 12, 2026

Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		3665.21 KPa.
Shell Thickness	ts	9.5300 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	600.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		2637.35 KPa.
Channel Thickness	tc	9.5300 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	600.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	583
Tube Wall Thickness	et	2.1080 mm
Tube Outside Diameter	D	15.8750 mm
Total Straight Tube Length	Lt	4877.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4757.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	19.8440 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	285.351 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	1839.8 cm ²



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ASME TS Calc: Pressure Test

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Perimeter of Tube Layout Cp 1715.8 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	60.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	740.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	663.00 mm
Step 1 Depth on the Tubeside	ht1	8.00 mm
Step 2 Diameter on the Tubeside	dt2	597.00 mm
Step 2 Depth on the Tubeside	ht2	8.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		663.00 mm

Note: Tubesheet diameter is now: 663.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes AL 0.0 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	21.11 °C
Mean Tube Metal Temp. along Tube length	Ttm	21.11 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	740.000 mm
Flange Inside Diameter	B	600.000 mm
Flange Face Outside Diameter	Fod	666.000 mm
Flange Face Inside Diameter	Fid	600.000 mm
Gasket Outside Diameter	Go	655.000 mm
Gasket Inside Diameter	Gi	629.000 mm
Small end Hub thk.	g0	9.5300 mm
Large end Hub thk.	g1	13.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	690.000 mm
Nominal Bolt Diameter	dB	19.0500 mm
Thread Series		TEMA Thread Series



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ASME TS Calc: Pressure Test

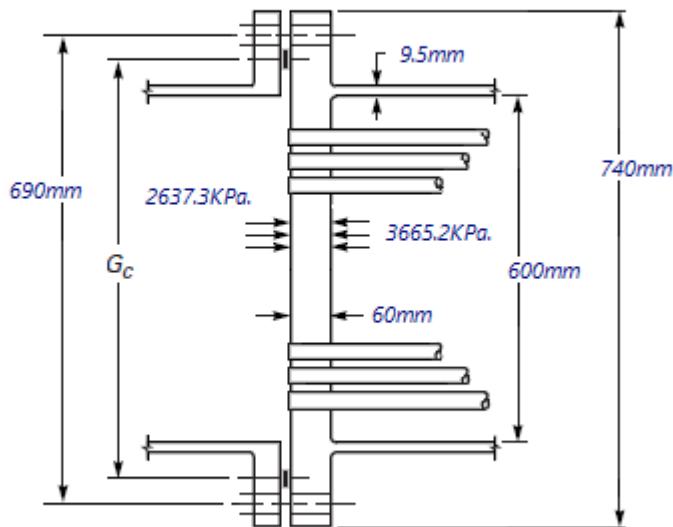
Case: 1 9:53a Aug 12, 2026

Number of Bolts	n	28
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with		Shell
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	44.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.



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Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
Please insure these properties are consistent with the
type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(2637.35 \cdot (15.875/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 2637.35)$
= $0.2240 + 0.0000 = 0.2240$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1080	15.88	4757.00	7.53	50.0000	0.0193957	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 19593.4844$ KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5412	15.88	4757.00	29.33	50.0000	0.0012784	80.65

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 80.6473) / (3 \cdot 29.3337) = 3665.5273$ KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5412 mm
Allowable Internal Pressure at Corroded thickness	27453.64 KPa.
Required Internal Design Pressure	2637.35 KPa.
Allowable External Pressure at Corroded thickness	19593.48 KPa.
Required External Design Pressure	3665.21 KPa.
Required Thickness due to Shell Side pressure	0.5412 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	642.156 mm

Warning: Available bolt area, 54.555 cm² for the flange is < required 61.585 cm²



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Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

= l_{tx} / h (modified for corrosion if present)

= 0/60.0

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter [d^*]:

= $\max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$

= $\max(15.875 - 2 \cdot 2.108 (0.20270E+09/.20270E+09) \cdot (92/137) \cdot (0); 15.875 - 2 \cdot 2.108)$

= 15.8750 mm

Equivalent outer tube limit circle diameter [Do]:

= $2 \cdot r_o + dt$

= $2 \cdot 285.351 + 15.875$

= 586.577 mm

Determine the basic ligament efficiency for shear [μ]:

= $(p - dt) / p$

= $(19.844 - 15.875) / 19.844$

= 0.200

Equivalent outer tube limit radius [ao]:

= $Do / 2$

= 586.577/2

= 293.2885 mm

Effective tube pitch [p^*]:

= $p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$

= $19.844 / \sqrt{ 1 - 4 \cdot \min(0 \cdot 100.0; 4 \cdot 586.577 \cdot 19.844) / (\pi \cdot 586.577^2) }$

= 19.8440 mm

Effective ligament efficiency for bending [μ^*]:

= $(p^* - d^*) / p^*$

= $(19.844 - 15.875) / 19.844$

= 0.2000

Ratio [ρ_s]:

= a_s / a_o

= 300.0/293.2885

= 1.0229

Ratio [ρ_c]:

= a_c / a_o

= 321.0778/293.2885

= 1.0948

Compute parameter [xt]:

= $1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$

= $1 - 583 ((15.875 - 2 \cdot 2.108) / (2 \cdot 293.2885))^2$

= 0.7697

Determine parameter [xs]:

= $1 - Nt(dt / (2 \cdot a_o))^2$

= $1 - 583 (15.875 / (2 \cdot 293.2885))^2$



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$$= 0.5730$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts)Es/L$$

$$= \pi \cdot 9.53 (600.0 + 9.53) \cdot 2.0270E+09 / 4757.0$$

$$= 79297.1719 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(D_t - tt)Et/L$$

$$= \pi \cdot 2.108 (15.875 - 2.108) \cdot 2.0270E+09 / 4757.0$$

$$= 396.1682 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 79297.172 / (583 \cdot 396.168)$$

$$= 0.34333$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_j)$$

$$= 1 / (1 + 79297.172/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - v_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((600.0 + 9.53) 9.53)$$

$$= 0.0239 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - v_s^2))$$

$$= 0.024 \cdot 0.20270E+09 \cdot 9.53^3 / (6(1 - 0.3^2))$$

$$= 766397568.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(600.0 + 9.53) \cdot 0.76640E+09 / 60.0^3 \cdot (1 + 60.0 \cdot 0.024 + (60.0 \cdot 0.024)^2/2)$$

$$= 44833552.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot v_s / (2(D_s + ts)))$$

$$= 600.0^2 / (4 \cdot 0.20270E+09 \cdot 9.53) \cdot (1 - 600.0 \cdot 0.3 / (2(600.0 + 9.53)))$$

$$= 0.39709E-01 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 3.023584 ; \mu^* = 0.200010 ; E^* = 31491868 \text{ KPa}$$

$$E^*/E = 0.155361 ; v^* = 0.458557$$

Note: As h/p (3.024) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - v^*)N_t \cdot E_t \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.459^2) 583 \cdot 0.20270E+09 \cdot 2.108$$

$$(15.875 - 2.108) 293.2885^2) / (31491868 \cdot 4757.0 \cdot 60.0^3))^{1/4}$$

$$= 3.6256$$



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Values from Table 4.18.3

$$\begin{aligned} Z_d &= 0.031494 ; Z_v = 0.073484 ; Z_m = 0.394465 \\ Z_a &= 0.455587E+01 ; Z_w = 0.073484 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$\begin{aligned} &= A / D_o \\ &= 663.0/586.577 \\ &= 1.1303 \end{aligned}$$

Compute coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.46) / 31491868 \cdot (44833552 + 0 + 0.20270E+09 \cdot \ln(1.13)) \\ &= 1.1976 \end{aligned}$$

Compute parameter [ϕ]:

$$\begin{aligned} &= (1 + v^*) F \\ &= (1 + 0.4586) 1.1976 \\ &= 1.7468 \end{aligned}$$

Compute parameter [Q1]:

$$\begin{aligned} &= (\psi_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m) \\ &= (1.0229 - 1 - 1.7468 \cdot 0.0735) / (1 + 1.7468 \cdot 0.3945) \\ &= -0.062449 \end{aligned}$$

Compute Parameter [Qz1]:

$$\begin{aligned} &= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4 \\ &= (0.03149 + -0.06245 \cdot 0.07348) / 2 \cdot 3.62562^4 \\ &= 2.3245 \end{aligned}$$

Compute parameter [Qz2]:

$$\begin{aligned} &= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4 \\ &= (0.07348 + -0.06245 \cdot 0.39447) / 2 \cdot 3.62562^4 \\ &= 4.2205 \end{aligned}$$

Compute parameter [U]:

$$\begin{aligned} &= (Z_w + (\psi_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m) \\ &= (0.0735 + (1.0229 - 1) 0.3945) 3.62562^4 / (1 + 1.7468 \cdot 0.3945) \\ &= 8.4410 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

$$\begin{aligned} &= (G_c - C) / D_o \text{ (config b)} \\ &= (642.1556 - 690.0) / 586.577 = -0.08157 \end{aligned}$$

Compute Parameter [γ]:

$$= 0.000 \text{ mm (For Pressure only cases)}$$

Calculate parameter [ω_s]:

$$\begin{aligned} &= \psi_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s) \\ &= 1.0229 \cdot 0.76640E+09 \cdot 0.0239 \cdot 0.039709 (1 + 60.0 \cdot 0.0239) \\ &= 1805.0923 \text{ mm}^2 \end{aligned}$$

Calculate parameter [ω^*s]:

$$\begin{aligned} &= a o^2 (\psi_s^2 - 1) \cdot (\psi_s - 1) / 4 - \omega_s \\ &= 293.289^2 (1.023^2 - 1) \cdot (1.023 - 1) / 4 - 1805.092 \\ &= -1782.3126 \text{ mm}^2 \end{aligned}$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta C (C_m \theta_{\text{Theta}} \cdot \beta_C + C_q \Delta \cdot h \cdot \beta_C^2) - \zeta_c (C_m \theta_{\text{Theta}} + h \cdot C_q \theta_{\text{Theta}} \cdot h \cdot \beta_C / 2)] \\ &= 1.095 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 60.0) - \end{aligned}$$



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$$0 (0/60.0 + 0 \cdot 0/2)) \\ = 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1)/4 - (ps - 1)/2] - \omega c \\ = 293.28851^2 [(1.09475^2 + 1) \cdot (1.09475 - 1)/4 - (1.02288 - 1)/2] - 0 \\ = 3495.3523 \text{ mm}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - \\ [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \} \\ = 3665.206 \{ 0.573 + 2(1 - 0.573) 0.3 + [2/0.343 \cdot \\ (600.0/586.577)^2] 0.3 - [(1.023^2 - 1)/(1.0 \cdot 0.343)] - \\ [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.343)] [(0^2 - (600.0)^2)/586.577^2] \} \\ = 9246.8027 \text{ KPa.}$$

Pressure [$P't$]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt \\ = [0.77 + 2(1 - 0.77) 0.3 + 1/(1. \cdot 0.343)] 2637.349 \\ = 10076.0820 \text{ KPa.}$$

Pressure [P_y]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2) \\ = 583 \cdot 396.168 \cdot 0 / (3.142 \cdot 293.289^2) \\ = 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot ao^2) \\ = -0.082 \cdot 8.441 \cdot 108010.19 / (2 \cdot \pi \cdot 293.289^2) \\ = 1349.2620 \text{ KPa.}$$

Calculate the pressure [$Prim$]:

$$= - (U/ao^2) \cdot (\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) \\ = - (8.441/293.289^2) \cdot (-1782.313 \cdot 3665.206 - 3495.352 \cdot 2637.349) \\ = 1545.6444 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/ao^2) \cdot (\omega \cdot s \cdot P's - \omega \cdot c \cdot P't) \\ = (8.441/293.289^2) \cdot (1805.0923 \cdot 0 - 0 \cdot 0) \\ = 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst (Qz1 + (ps - 1) Qz2)) \cdot \\ (P's - P't + P_y + P_w + Pw + Prim) \\ = 0.1000E+01 \cdot 0.343 / (1 + 1.0 \cdot 0.343 (2.324 + (1.023 - \\ 1) 4.22)) \cdot (9246.803 - 10076.082 + 0 + 0 + 1349.262 + 1545.644) \\ = 387.2765 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot Ps - \omega \cdot c \cdot Pt) - (\omega \cdot s \cdot P's - \omega \cdot c \cdot P't)) + \\ W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm) \\ = [((-1782.313 \cdot 3665.206 - 3495.352 \cdot 2637.349) - \\ (1805.092 \cdot 0 - 0 \cdot 0)) \cdot 0. + \\ 108010.2 \cdot -0.082 / (2 \cdot \pi)] / (1 + 1.74684 \cdot 0.39447) \\ = -1781.0778 \text{ Kg}$$



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Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.062 + 2 \cdot -1781.078 / (387.276 \cdot 293.289^2)$$

$$= -1.11104$$

Fm Value from Table 4.18.3 = 0.555519

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.5555 / 0.2) \cdot (2 \cdot 293.2885 / (60.0 - 0))^2 \cdot 387.28$$

$$= 154.2170 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 250.0 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ]:

$$= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.2607 / 0.2) \cdot (2 \cdot 293.2885 / (35.927 - 0))^2 \cdot 683.59$$

$$= 356.2401 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 35.9266 + 0 + 0$$

$$= 35.9266 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.2 \cdot 60.0 / 293.289) \cdot 189.997$$

$$= 15.5483 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [Pe] <= TauIimit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\text{Tubesheet thickness (Incl. Corr.)} = 3.8100 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 65.7634 \text{ N./mm}^2$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(HreqB; HreqS)$$

$$= \max(35.9266; 3.81)$$

$$= 35.9266 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = -4.3328, Ftmax = 3.498$$

First Extreme Tube Axial Stress from among all the tubes [σ t1]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs)$$

$$= ((3665.21 \cdot 0.573 - 2637.35 \cdot 0.7697) - (387.276) \cdot -4.333) /$$

$$(0.7697 - 0.573)$$

$$= 8.8884 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ t2]:

$$= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs)$$

$$= ((3665.21 \cdot 0.573 - 2637.35 \cdot 0.7697) - (387.276) \cdot 3.498) /$$

$$(0.7697 - 0.573)$$

$$= -6.5308 \text{ N./mm}^2$$

Maximum Tube Axial Stress [σ t,max]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 8.888 \text{ N./mm}^2$$



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ASME TS Calc: Pressure Test

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Allowable Tube Stress [σ_t, A]:
 $= \beta T \cdot S_{y,t} = 170.9969 \text{ N./mm}^2$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:
 $= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w) X_{a^4}); 1.25)$
 $= \max((3.25 - 0.25(0.031 + -1.111 \cdot 0.073) 3.626^4); 1.25)$
 $= 1.2500 \text{ (Should be } \leq 2 \text{)}$

Determine the factor [rt]:
 $= (\sqrt{ dt^2 + (dt - 2 \cdot tt)^2 }) / 4$
 $= (\sqrt{ 15.875^2 + (15.875 - 2 \cdot 2.108)^2 }) / 4$
 $= 4.9241 \text{ mm}$

Determine the factor [Ct]:
 $= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{yt})}$
 $= \sqrt{ (2 \cdot \pi^2 \cdot 0.20270E+09 / 179)}$
 $= 156.4208$

Determine the factor [Ft]:
 $= k \cdot L / r$
 $= 0 \cdot 0 / 4.924$
 $= 0.0000$

Buckling Allowable Stress [Stb]:
 $= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$
 $= (179 / 1.25) \cdot (1 - 0 / (2 \cdot 156.421))$
 $= 170.997 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:
 $= \sigma_{t,max} \cdot \text{Tube Area}$
 $= 8.89 \cdot 0.9117$
 $= 82.63 \text{ Kg}$

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (15.875 - 2.108) 2.108$
 $= 0.912 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.108 (15.875 - 2.108) 170.997$
 $= 1589.716 \text{ Kg}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$
 $= 0.85 \cdot \pi \cdot 2.0 (15.875 + 0.67 \cdot 2.0) 171.0$
 $= 1589.7156 \text{ Kg}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 170.997 / 170.997$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$
 $= 0.55 \cdot \pi \cdot 2.0 (15.875 + 0.67 \cdot 2.0) 171.0$



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= 1037.3115 Kgf

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd) - 0.75 \cdot do} \right) - 0.75 \cdot 15.875$$

$$= 2 \left(\sqrt{((0.75 \cdot 15.875)^2 + 1.07 \cdot 2.108 \cdot (15.875 - 2.108) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 15.875 \right)$$

$$= 2.479 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2648 mm

Minimum Required Groove Weld Leg [agr]: 1.2648 mm

Tube-Tubesheet Jt allowable, 1589.72 is >= tube strength 1589.72 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot S_{y,s} = 0.95 \cdot 240.0 = 341.99 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 293.289^2 / ((600.0 + 9.53) 9.53) \cdot [387.28 + (1.023^2 - 1)(3665.21 - 2637.35)] + 300.0^2 \cdot 2637.35 / ((600.0 + 9.53) 9.53)$$

$$= 47.3044 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 9.53 mm, the minimum Shell length adjacent to the tubesheet should be 136.111 mm

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	154.2 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	3.5 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	8.9 <=	171.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-6.5 <=	-171.0 N./mm ²	Pass
Maximum Force on any one Tube	82.6 <=	1589.7 Kgf	Pass
Axial Membrane Stress in Shell	47.3 <=	342.0 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	35.927	60.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.265	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.265	2.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd	Shear Allwd	Case Type	Pass/Fail
D1uc	27.275	17.768	133	356	13	190 Ps+Pt-Th D1 Ok
D2uc	19.196	...	74	356	17	190 Ps+Pt-Th D2 Ok
D3uc	35.927	...	154	356	3	190 Ps+Pt-Th D3 Ok



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ASME TS Calc: Pressure Test

Case: 1 9:53a Aug 12,2026

Max: 35.9266 17.768 mm 0.433 0.087 (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:		Fail
D1uc	17	341	...	...	:	...	...	...	...	:		Ok
D2uc	30	341	...	...	:	...	...	...	...	:		Ok
D3uc	47	341	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.138			...	:		...		...	:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	19	170	-19	-170	177	1589	...	341	...	...	Ok
D2uc	13	170	-8	-170	117	1589	...	341	...	...	Ok
D3uc	9	170	-7	-170	83	1589	...	341	...	...	Ok
Max RATIO	0.112			0.112		0.112		...		...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	35.927	60.000	Ok
Tubesheet thickness flanged extension :	17.768	44.000	Ok
Tube thickness :	0.541	2.108	Ok
Tube-tubesheet fillet weld leg :	1.265	2.000	Ok
Tube-tubesheet groove weld leg :	1.265	2.000	Ok

Min Shell length of thk, (9.530) adj. to tubesheet: 136.111 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
Due to Pressure Shell Compresses by : -0.090 mm



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ASME TS Calc: Pressure Test

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Due to Pressure + Thermal Shell Compresses by : -0.090 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0118	...	...	...	...	...
2	-0.0896	...	...	...	...	...
3	-0.0777	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	616.68	5.165	0.0011633	89.8136
20	30	726.2	619.06	6.355	0.0011829	90.2975
30	40	No Calc	...	61.825	No Calc	No Calc
40	50	4790	619.06	6.355	0.00015459	15.456
50	60	No Calc	...	61.825	No Calc	No Calc
60	70	726.2	619.06	6.355	0.0011829	90.2975
70	80	No Calc	616.68	5.165	0.0011633	89.8136

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	8.34	No Calc	...	835.766
20	30	9.53	No Calc	...	1235.87
30	40	65	54.8386	...	No Calc
40	50	9.53	No Calc	...	211.541
50	60	65	54.8386	...	No Calc
60	70	9.53	No Calc	...	1235.87
70	80	8.34	No Calc	...	835.766

Minimum

212

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	726.2	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	4790	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	726.2	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 13

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	43.6214	49828.3	29.0886	51194.2	4.36214
20	30	84.863	169676	56.8849	173287	8.4863
30	40	80.0512	23683.1	76.2136	23769.2	8.00512
40	50	677.49	801743	454.132	830567	67.749
50	60	80.0512	23683.1	76.2136	23769.2	8.00512
60	70	84.863	169676	56.8849	173287	8.4863
70	80	43.6214	49828.3	29.0886	51194.2	4.36214
Total		1094	1288119.12	778	1327067.88	109



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	49.7979	-50	...	...	Liquid: 10
20	Liqd	169.573	300	...	...	Liquid: 30
20	Nozl	29.7678	300	-384.137	...	N1- Channel inle
40	Sadl	82.383	960	419.235	...	Left Saddle
40	Sadl	82.383	3830	419.235	...	Right saddle
40	Liqd	40.0627	2395	...	...	Liquid: 40
40	Insl	67.5312	2395	...	...	Hot insulation
40	Nozl	86.7305	500	436.525	...	N1- Shell outlet
40	Nozl	121.14	4000	461.925	...	N2- shell inlet
60	Liqd	169.573	300	...	...	Liquid: 50
60	Nozl	29.7678	300	384.137	...	N2- Channel outl
70	Liqd	49.7979	126.2	...	...	Liquid: 70
30	FTsh	160.974	108.8	...	...	Fixed tubesheet
30	Tube	2210.06	2517.3	...	...	Tubes
30	RTsh	160.974	4925.8	...	...	Tubesheet
40	Bafl	5.14714	596	...	...	Baffles
40	Bafl	5.14714	1192	...	...	Baffles
40	Bafl	5.14714	1788	...	...	Baffles
40	Bafl	5.14714	2384	...	...	Baffles
40	Bafl	5.14714	2980	...	...	Baffles
40	Bafl	5.14714	3576	...	...	Baffles
40	Bafl	5.14714	4172	...	...	Baffles
40	Bafl	5.14714	4768	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	164.8
Liquid	478.8
Insulation	67.5
Nozzles	267.4
Exchanger Components	2573.2
Liquid in Tubes	303.4

Sum of the Detail Weights 3855.1 kg.

Weight Summation Results: (kg.)



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Element and Detail Weights:

Step: 13 9:53am Aug 12,2026

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1204.0	1204.0	1204.0	1204.0	1204.0	1204.0
Saddles	164.8	164.8	164.8	164.8	164.8	164.8
Nozzles	267.4	267.4	267.4	267.4	267.4	267.4
Exchanger	2573.2	2573.2	2573.2	2573.2	2573.2	2573.2
Insulation	...	...	67.5	67.5	67.5	67.5
Ope. Liquid	...	...	...	...	...	478.8
Tube Ope Lqd	...	...	...	...	...	303.4
Test Liquid	...	1287.3	...	...	...	...
Tube Tst Lqd	...	303.4	...	...	...	...
Totals	4209.4	5800.1	4276.9	4276.9	4276.9	5059.1

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	4209.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	5800.1 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	4276.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	4276.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	4276.9 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	5059.1 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	4711.5 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	303606.8 cm ³
Weight of Ope Liq in tubes :	303.4 kg.
Weight of Water in tubes :	303.4 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	5636.18	5338.68
20	30	11669	11309.7
30	40	3335.49	1551.32
40	50	93157.6	90289.4
50	60	3335.49	1551.32
60	70	11669	11309.7
70	80	5636.18	5338.68

Total	134438.9	126688.9 cm ²
Total	144.7	136.4 ft ²

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Nozzle Flange MAWP:

Step: 14 9:53am Aug 12,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N1- Channel in	4880.9	5100.0	66	300	GR 1.1	135.5	4880.9	4880.9	4880.9
N1- Shell outl	4469.6	5100.0	166	300	GR 1.1	95.6	4469.6	4469.6	4469.6
N2- shell inle	4469.6	5100.0	166	300	GR 1.1	82.8	4469.6	4469.6	4469.6
N2- Channel ou	4880.9	5100.0	66	300	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4469.575 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 4880.867 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 5100.005 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1- Channel inle	196.088	0.50	0	408	2028.7	4880.9	Pass
N1- Shell outlet	302.387	0.50	1	683	2819.4	4469.6	Pass
N2- shell inlet	357.251	0.50	2	826	2819.4	4469.6	Pass

Evaluating Flange on Nozzle [N1- Channel inle]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(0) + 4 \cdot \text{abs}(408)196.088 \leq \pi \cdot 196.088^3 \cdot ((4880.867 - 2028.730) + .500 \cdot 4880.867)$$

$$= 0.32729E+03 \leq 0.12784E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1- Channel inle [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4880.867, 0.500 \cdot 4880.867 + 4880.867 - 327.3 / (\pi \cdot 196.088^3))$$

$$= \min(4880.867, 7185.803)$$

$$= 4880.867 \text{ KPa.}$$

Evaluating Flange on Nozzle [N1- Shell outlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1) + 4 \cdot \text{abs}(683)302.387 \leq \pi \cdot 302.387^3 \cdot ((4469.575 - 2819.390) + .500 \cdot 4469.575)$$

$$= 0.84683E+03 \leq 0.34414E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1- Shell outlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4469.575, 0.500 \cdot 4469.575 + 4469.575 - 846.8 / (\pi \cdot 302.387^3))$$

$$= \min(4469.575, 6608.764)$$

$$= 4469.575 \text{ KPa.}$$

Evaluating Flange on Nozzle [N2- shell inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$



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Nozzle Flange MAWP: Step: 14 9:53am Aug 12, 2026

$$\begin{aligned} &= 16(2) + 4 \cdot \text{abs}(826) 357.251 \leq \pi \cdot 357.251^3 [(4469.575 - 2819.390) + .500 \cdot 4469.575] \\ &= 0.12098\text{E}+04 \leq 0.56750\text{E}+05 \quad \text{Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2- shell inlet [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4469.575, 0.500 \cdot 4469.575 + 4469.575 - 1209.8 / (\pi \cdot 357.251^3)) \\ &= \min(4469.575, 6621.543) \\ &= 4469.575 \text{ KPa.} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation:

Step: 15 9:53am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	
Use Gust Response Factor (Dynamic Analysis)	No	

Calculated Values:

Vessel Length to Diameter ratio: 9.787

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)

= 1983.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.1 • 0.896 • 1983.503

= 236.4 N

From	Height mm	k1	k2	k3	Vz m/sec	Cf
10	860.00	1.0000	1.0000	1.0000	57.50	0.8957
20	860.00	1.0000	1.0000	1.0000	57.50	0.8957
30	860.00	1.0000	1.0000	1.0000	57.50	0.8957
40	860.00	1.0000	1.0000	1.0000	57.50	0.8957
50	860.00	1.0000	1.0000	1.0000	57.50	0.8957
60	860.00	1.0000	1.0000	1.0000	57.50	0.8957
70	860.00	1.0000	1.0000	1.0000	57.50	0.8957

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm²	Wind Pressure Kgs/m²	Element Wind Load Kgf
10	20	860	740.016	1330.8	202.347	26.5234
20	30	860	742.872	4457.23	202.347	88.8347
30	40	860	720	570.96	202.347	11.3795
40	50	860	850.872	40756.8	202.347	812.302
50	60	860	720	570.96	202.347	11.3795
60	70	860	742.872	4457.23	202.347	88.8347
70	80	860	740.016	1330.8	202.347	26.5234



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Wind Load Calculation:

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The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation: Step: 16 9:53am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User	I	2.0000
Zone Number		2
Soil Type	Rocky or Hard Soil	
Response reduction factor (Force factor)	R	2.0000
Time Period of Vibration (program or user)		0.0303 Seconds
Value of Sa/g from user Spectrum Data		0.0939
Percent Damping		2.0 %
Considering MCE for load determination		No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:

= (Sa/g)ss • DampVal) • (I / R)

= (0.0939 • 1.0) • 1.0

= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs

Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.

The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:

= Earthquake Weight • Ah • Scalar

= 505.19 • 0.0939 • 1.0

= 47.4 Kgf

Vertical load per 8.4:

= 2/3 • Ah • W • Scalar

= 2/3 • 0.0939 • 4547 • 1.0



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Earthquake Load Calculation:

Step: 16

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= 284.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	300	505.189	47.4151
20	30	300	505.189	47.4151
30	40	300	505.189	47.4151
40	Sad1	300	505.189	47.4151
Sad1	50	300	505.189	47.4151
40	50	300	505.189	47.4151
50	60	300	505.189	47.4151
60	70	300	505.189	47.4151
70	80	300	505.189	47.4151

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 17 9:53am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3215.000 mm
Center of Gravity of the Liquid	3214.999 mm
Center of Gravity of the Insulation	3215.000 mm
Center of Gravity of the Nozzles	3327.469 mm
Center of Gravity of the Tubesheet(s)	3193.500 mm
Center of Gravity of the Tubes	3193.500 mm
Center of Gravity of the Baffles	3501.999 mm
Center of Gravity of Bare Shell New and Cold	3215.000 mm
Center of Gravity of Bare Shell Corroded	3215.000 mm
Vessel CG in the Operating Condition	3210.952 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	3212.067 mm
Vessel CG in the Test Condition	3211.750 mm

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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

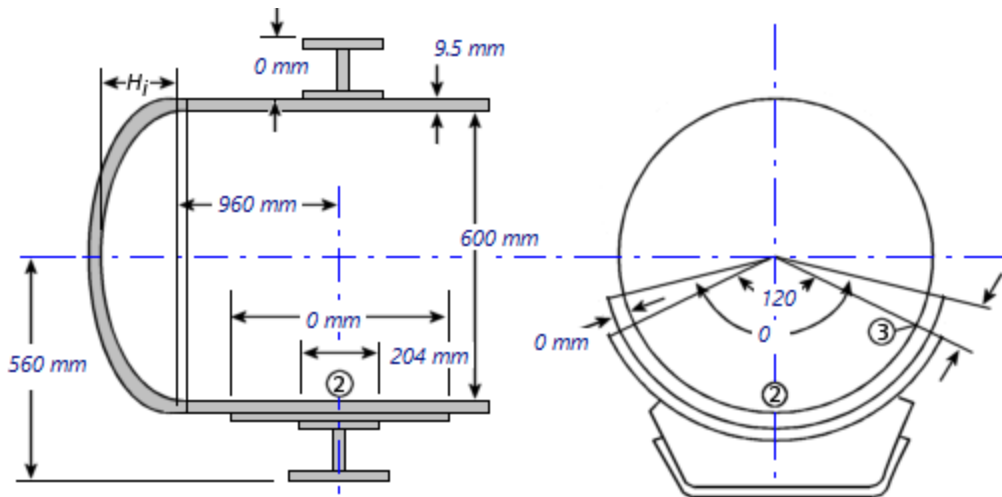
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{306.3525 \cdot 6.355}, 2 \cdot 960.0)$$

$$= 272.8324 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	306.35	mm
Shell Thickness used in this Case	t	9.530	mm
Stiffened Vessel Length per 4.15.6	L	4790.00	mm
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		140.75	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	560.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	4049.97	Kgf
Pressure used in Analysis	P	2600.147 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	57.43	140.75
Long. Stress at Bottom of Midspan	67.92	140.75
Long. Stress at Top of Saddles	98.98	140.75
Long. Stress at Bottom of Saddles	42.55	140.75



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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12,2026

Tangential Shear in Shell	14.31	112.60
Circ. Stress at Horn of Saddle	83.69	175.94
Circ. Compressive Stress in Shell	1.74	140.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(1065.8/2 + 0) \cdot 560.0/550.0$$

$$= 1627.7 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(96.47, 0) \cdot 560.0/2870.0$$

$$= 18.8 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$$

$$= \max(426.74, 0) \cdot 560.0/2870.0$$

$$= 83.3 \text{ Kg}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(427/2 + 0) \cdot 560.0/550.0$$

$$= 651.7 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2422 + \max(19, 1628, 83, 652)$$

$$= 4050.0 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.619 \cdot 1.2)^2)$$

$$= 946.068 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4132.36	Kgf
Transverse Shear Load	532.89	Kgf
Longitudinal Shear Load	426.74	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 550.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)]$$

$$= -4050 \cdot 960.0 [1 - (1 - 960.0/4790.0 + (306.352^2 - 0.0^2)/(2 \cdot 960.0 \cdot 4790.0))/(1 + (4 \cdot 0.0)/(3 \cdot 4790.0))]$$

$$= -739.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 4050 \cdot 4790/4((1 + 2(306^2 - 0^2)/(4790^2))/(1 + (4 \cdot 0)/$$



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$$(3 \cdot 4790) - 4 \cdot 960 / 4790 \\ = 1001.6 \text{ Kg-m.}$$

$$\text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ = P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 2600.147 \cdot 306.352 / (2 \cdot 6.355) - 1001.6 / (\pi \cdot 306.4^2 \cdot 6.355) \\ = 57.43 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ = P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 2600.147 \cdot 306.352 / (2 \cdot 6.355) + 1001.6 / (\pi \cdot 306.4^2 \cdot 6.355) \\ = 67.92 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Top of Shell at Support (4.15.8) } [\sigma^*3]: \\ = P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 2600.147 \cdot 306.352 / (2 \cdot 6.355) - 739.6 / (0.1066 \cdot \pi \cdot 306.4^2 \cdot 6.355) \\ = 98.98 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) } [\sigma^*4]: \\ = P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 2600.147 \cdot 306.352 / (2 \cdot 6.355) + 739.6 / (0.1923 \cdot \pi \cdot 306.4^2 \cdot 6.355) \\ = 42.55 \text{ N./mm}^2$$

$$\text{Maximum Shear Force in the Saddle (4.15.3) } [T]: \\ = Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 4050(4790.0 - 2 \cdot 960.0) / (4790.0 + (4 \cdot 0.0 / 3)) \\ = 2426.6 \text{ Kg}$$

$$\text{Shear Stress in the shell no rings, not stiffened (4.15.12) } [\tau_2]: \\ = K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 2426.6 / (306.3525 \cdot 6.355) \\ = 14.31 \text{ N./mm}^2$$

$$\text{Decay Length (4.15.20) } [x_1, x_2]: \\ = 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{306.352 \cdot 6.355} \\ = 34.416 \text{ mm}$$

$$\text{Circumferential Stress in shell, no rings (4.15.21) } [\sigma_6]: \\ = -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2)) \\ = -0.7603 \cdot 4050 \cdot 0.1 / (6.355(204.0 + 34.42 + 34.42)) \\ = -1.74 \text{ N./mm}^2$$

$$\text{Circ. Comp. Stress at Horn of Saddle, } L \geq 8R_m \text{ (4.15.22) } [\sigma_7]: \\ = -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2) \\ = -4050 / (4 \cdot 6.355(204.0 + 34.416 + 34.416)) - \\ 3 \cdot 0.0529 \cdot 4050 / (2 \cdot 6.355^2) \\ = -83.69 \text{ N./mm}^2$$

$$\text{Effective reinforcing plate width (4.15.24) } [B_1]: \\ = \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(204.0 + 1.56 \cdot \sqrt{306.352 \cdot 6.355}, 2 \cdot 960.0) \\ = 272.83 \text{ mm}$$

$$\text{Distance between Saddle Supports } [L_s]: \\ = 2870.0 \text{ mm}$$

$$\text{Free Un-Restrained Thermal Expansion between the Saddles } [\text{Exp}]: \\ = \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000013 \cdot 2870.0 (165.6 - 21.1) \\ = 5.193 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	550.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm



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Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(303.175 \cdot 6.355)} + 225.0)6.355^3)/12$$

$$= 0.628 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 0.628 + 18.65(104.303 - 3.177)^2$$

$$= 1907.877 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	293.5	6.4	3.2	18.7	5926.1	0.191E+04
Wearplate	225.0	12.0	12.4	27.0	33358.5	0.229E+04
Web	16.0	222.5	129.6	35.6	461278.3	0.170E+04
BasePlate	150.0	16.0	248.8	24.0	597180.1	0.502E+04
Totals	...	...	...	105.2	1097743.0	0.109E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1097743/105.246$$

$$= 104.303 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4049.972$$

$$= 824.2572 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	0.9335	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	157.1972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 560.0 - 303.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 309.276 \text{ mm}$$

Bending Moment, M = Fh * d	=	254.9283	Kg-m.
----------------------------	---	----------	-------

Bending Stress, Sb = (M * C1 / I)	=	2.3906	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	174.6635	N./mm ²

Minimum Thickness of Baseplate per Moss:



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$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(4050 + 82)150.0 / (2 \cdot 550.0 \cdot 174.663)}$$

$$= 9.743 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 556.9062 / (3 - 1)$$

$$= 278.453 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 278.4531 \cdot 150.0 / 2$$

$$= 208.840 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4132.355 / (550.0 \cdot 150.0)$$

$$= 5.009 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 208.8 \cdot 5.01$$

$$= 1046.061 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 30.08 + 22.276$$

$$= 52.356 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 1046.1 / 52.3563$$

$$= 1.959 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.145E+04
Web	139.2	16.0	8.0	22.3	17821.0	770.
Totals	...	...	...	52.4	348700.9	0.222E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204.0 - 16.0$$

$$= 188.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 348700.938 / 52.356$$

$$= 66.602 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(66.602, 204.0 - 66.602)$$

$$= 137.398 \text{ mm}$$



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Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2222.2/52.356}$$

$$= 65.150 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 560.0 - \cos(120.0/2) (300.0 + 9.53 + 12.0) - 16.0$$

$$= 383.235 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 550.0 - 50.8$$

$$= 499.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (556.906 - 499.2) / 2$$

$$= 28.853 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{383.235^2 + 28.853^2}$$

$$= 384.320 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.20270\text{E}+09 / 262.0}$$

$$= 123.583$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 384.32 / 65.15$$

$$= 5.899$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 426.7 / (2 \cdot 550.0) \cdot 278.453 \cdot 384.32 / 2$$

$$= 20.758 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($5.899 < 123.583$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (5.9)^2 / (2 \cdot 123.58^2)) 262 / (5/3 + 3 \cdot (5.9) / (8 \cdot 123.58) - (5.9)^3 / (8 \cdot 123.58^3))$$

$$= 155.4 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.96 / 155.35 + (20.76 \cdot 137.398 / 22222406) / 174.66$$

$$= 0.020$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.200E+04
Web	278.5	16.0	8.0	44.6	35642.0	762.
Totals	...	...	...	74.6	366522.0	0.276E+04



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 366521.969/74.632$$

$$= 49.110 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(49.11, \text{abs}(204.0 - 49.11))$$

$$= 154.890 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 560.0 - \cos(321.53/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 222.470 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (2763.7/74.632) }$$

$$= 60.852 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 222.47/60.852$$

$$= 3.656$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 426.736/(2 \cdot 550.0)$$

$$= 0.388 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.388 \cdot 278.453 \cdot 222.47$$

$$= 24.032 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 24.032/178427.25$$

$$= 1.321 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.6559 < 123.583$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.66)^2 / (2 \cdot 123.58^2)) 262 / (5/3 + 3 \cdot (3.66) / (8 \cdot 123.58) - (3.66^3) / (8 \cdot 123.58^3))$$

$$= 156.1 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 3.31/156.09 + (24.03 \cdot 154.89/2763.653) / 174.66$$

$$= 0.029$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	2.2586 cm ²



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Saddle Load QO (Weight)	QO	2504.6 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	83.3 Kgf
Maximum Transverse Force	Ft	532.9 Kgf
Maximum Longitudinal Force	F1	426.7 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
= 1245.7 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $\max(F_f, F1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $1245.7 / (2.259 \cdot 2)$
= 27.0 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $532.9 / (2.259 \cdot 2)$
= 11.6 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $426.74 / (107.87 \cdot 2.0)$
= 0.1940 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $560.0 \cdot 532.89 + 0.0$
= 298.42 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $298.42 / 2504.62$
= 119.15 mm > Bplen/6 --> Uplift in Transverse direction

f = $B_{plen} / 2 - \text{Edgedis}$
= $550.0 / 2 - 50.01$
= 224.99 mm

Modular Ratio Of Steel/Concrete (n1):
= E_S / E_C
= $203402.5 / 21526.32$
= 9.45

K1 = $3 \cdot (e - 0.5 \cdot B_{plen})$
= $3 \cdot (119.15 - 0.5 \cdot 550.0)$
= -467.56 mm

K2 = $6 \cdot n1 \cdot A_t / B_{pwid} \cdot (f + e)$
= $6 \cdot 9.45 \cdot 2.26 / 150.0 \cdot (224.99 + 119.15)$
= 29376.81 mm²

K3 = $-K2 \cdot (0.5 \cdot B_{plen} + f)$
= $-29376.81 \cdot (0.5 \cdot 550.0 + 224.99)$
= -14688033.34 mm³

Iteratively Solving for the Effective Bearing Length:

$$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$$

$$Y^3 + -467.56 \cdot Y^2 + 29376.81 \cdot Y + -0.1E+08 = 0$$



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$$Y = 471.35 \text{ mm}$$

$$\begin{aligned} \text{Num} &= (\text{Bplen} / 2 - Y / 3 - e) \\ &= (550.0 / 2 - 471.35 / 3 - 119.15) \\ &= -1.26 \end{aligned}$$

$$\begin{aligned} \text{Denom} &= (\text{Bplen} / 2 - Y / 3 + f) \\ &= (550.0 / 2 - 471.35 / 3 + 224.99) \\ &= 342.87 \end{aligned}$$

$$\begin{aligned} \text{Total Bolt Tension Force [Tforce]:} \\ &= -\text{QO} \cdot \text{Num} / \text{Denom} \\ &= -2504.62 \cdot -1.26 / 342.87 \\ &= 9.22 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= 9.22 / (137.29 \cdot 1.0) \\ &= 0.0066 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0.0, 0.194, 0.0066] \\ &= 0.1940 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned} \text{Bearing Pressure (fc)} \\ &= 2(\text{QO} + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\ &= 2(2504.62 + 9.22) / (471.35 \cdot 150.0) \\ &= 697.33 \text{ KPa.} \end{aligned}$$

$$\begin{aligned} \text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (\text{Bplen} - \text{WebIngh}) / 2 \\ &= (550.0 - 499.2) / 2 \\ &= 25.4000 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= \text{Tforce} \cdot \text{Adist} \\ &= 9.22 \cdot 25.4 \\ &= 0.23 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} \text{Equivalent Bearing Pressure (f1):} \\ &= \text{fc} \cdot (Y - \text{Adist}) / Y \\ &= 697.33 \cdot (471.35 - 25.4) / 471.35 \\ &= 659.75 \text{ KPa.} \end{aligned}$$

$$\begin{aligned} \text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (f1 + 2 \cdot \text{fc}) \\ &= (25.4^2 \cdot 150.0 / 6) \cdot (659.75 + 2 \cdot 697.33) \\ &= 3.38 \text{ Kg-m.} \end{aligned}$$

$$\begin{aligned} \text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(0.23, 3.38) / (150.0 \cdot 207.0))} \\ &= 2.5305 \text{ mm} \end{aligned}$$

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 9.53, tr = 5.366, c = 3.175 mm, E* = 1.0
Thickness Ratio = tr * E* / (tg - c) = 0.844, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B

-29 °C



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Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204.0 + 1.56 \sqrt{306.3525 \cdot 6.355}, 2 \cdot 960.0)$$

$$= 272.8324 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	306.35	mm
Shell Thickness used in this Case	t	9.530	mm
Stiffened Vessel Length per 4.15.6	L	4790.00	mm
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		140.75	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	560.00	mm
Coefficient of Friction	mu	0.50	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		4036.68	Kgf
Pressure used in Analysis	P	2600.147	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	57.45	140.75
Long. Stress at Bottom of Midspan	67.90	140.75
Long. Stress at Top of Saddles	98.86	140.75
Long. Stress at Bottom of Saddles	42.62	140.75
Tangential Shear in Shell	14.26	112.60
Circ. Stress at Horn of Saddle	83.42	175.94
Circ. Compressive Stress in Shell	1.74	140.75

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (1065.8/2 + 0) \cdot 560.0/550.0$$

$$= 1627.7 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(96.47, 0) \cdot 560.0/2870.0$$

$$= 18.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(426.74, 0) \cdot 560.0/2870.0$$

$$= 83.3 \text{ Kgf}$$



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Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(427/2 + 0) \cdot 560.0/550.0$$

$$= 651.7 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 2409 + \max(19, 1628, 83, 652)$$

$$= 4036.7 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.619 \cdot 1.2)^2)$$

$$= 946.068 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4119.07	Kgf
Transverse Shear Load	532.89	Kgf
Longitudinal Shear Load	426.74	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 550.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -4037 \cdot 960.0 [1 - (1 - 960.0/4790.0 + (306.352^2 - 0.0^2) /$$

$$(2 \cdot 960.0 \cdot 4790.0) / (1 + (4 \cdot 0.0) / (3 \cdot 4790.0))]$$

$$= -737.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 4037 \cdot 4790 / 4 ((1 + 2(306^2 - 0^2)/(4790^2)) / (1 + (4 \cdot 0) /$$

$$(3 \cdot 4790)) - 4 \cdot 960 / 4790)$$

$$= 998.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 2600.147 \cdot 306.352 / (2 \cdot 6.355) - 998.3 / (\pi \cdot 306.4^2 \cdot 6.355)$$

$$= 57.45 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 2600.147 \cdot 306.352 / (2 \cdot 6.355) + 998.3 / (\pi \cdot 306.4^2 \cdot 6.355)$$

$$= 67.90 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 2600.147 \cdot 306.352 / (2 \cdot 6.355) - 737.1 / (0.1066 \cdot \pi \cdot 306.4^2 \cdot 6.355)$$

$$= 98.86 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2600.147 \cdot 306.352 / (2 \cdot 6.355) + 737.1 / (0.1923 \cdot \pi \cdot 306.4^2 \cdot 6.355)$$

$$= 42.62 \text{ N./mm}^2$$



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Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a)/(L+(4 \cdot hm/3))$$

$$= 4037(4790.0 - 2 \cdot 960.0)/(4790.0 + (4 \cdot 0.0/3))$$

$$= 2418.6 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2418.64 / (306.3525 \cdot 6.355)$$

$$= 14.26 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(306.352 \cdot 6.355)}$$

$$= 34.416 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4037 \cdot 0.1 / (6.355(204.0 + 34.42 + 34.42))$$

$$= -1.74 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q/(4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -4037 / (4 \cdot 6.355(204.0 + 34.416 + 34.416)) - 3 \cdot 0.0529 \cdot 4037 / (2 \cdot 6.355^2)$$

$$= -83.42 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(306.352 \cdot 6.355)}, 2 \cdot 960.0)$$

$$= 272.83 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	550.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(303.175 \cdot 6.355)} + 225.0)6.355^3)/12$$

$$= 0.628 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 0.628 + 18.65(104.303 - 3.177)^2$$

$$= 1907.877 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	293.5	6.4	3.2	18.7	5926.1	0.191E+04
Wearplate	225.0	12.0	12.4	27.0	33358.5	0.229E+04
Web	16.0	222.5	129.6	35.6	461278.3	0.170E+04
BasePlate	150.0	16.0	248.8	24.0	597180.1	0.502E+04



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Totals	...	...	...	105.2	1097743.0	0.109E+05
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 1097743/105.246$$

$$= 104.303 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4036.684$$

$$= 821.5527 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.9304 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 560.0 - 303.17 \cdot \sin(120.0/2) / 1.0472$$

$$= 309.276 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 254.0918 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.3828 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(4037 + 82)150.0) / (2 \cdot 550.0 \cdot 174.663))}$$

$$= 9.727 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 556.9062 / (3 - 1)$$

$$= 278.453 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 278.4531 \cdot 150.0/2$$

$$= 208.840 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4119.067 / (550.0 \cdot 150.0)$$

$$= 4.993 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 208.8 \cdot 4.99$$



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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12,2026

= 1042.698 Kgf

Area of the Rib and Web [Ar]:

= Rib Area + Web Area

= 30.08 + 22.276

= 52.356 cm²

Compressive Stress [Sc]:

= P/Ar

= 1042.7/52.3563

= 1.953 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.145E+04
Web	139.2	16.0	8.0	22.3	17821.0	770.
Totals	...	...	...	52.4	348700.9	0.222E+04

Rib dimension [D]:

= Saddle Width - Web Thickness

= 204.0 - 16.0

= 188.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 348700.938/52.356

= 66.602 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)

= max(66.602, 204.0 - 66.602)

= 137.398 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{2222.2/52.356}$

= 65.150 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 560.0 - cos(120.0/2)(300.0 + 9.53 + 12.0) - 16.0

= 383.235 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(120.0/2)(300.0 + 9.53 + 12.0)

= 556.906 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 550.0 - 50.8

= 499.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (556.906 - 499.2)/2

= 28.853 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{383.235^2 + 28.853^2}$



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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12,2026

= 384.320 mm

Intermediate Term [Cc]:

$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

$= \sqrt{(2 \cdot \pi^2 \cdot 0.20270\text{E}+09 / 262.0)}$

= 123.583

Slenderness ratio [KL/r]:

= KL/r

= 1 • 384.32/65.15

= 5.899

Bending Moment [Rm]:

$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$

$= 426.7 / (2 \cdot 550.0) \cdot 278.453 \cdot 384.32 / 2$

= 20.758 Kg-m.

Compressive Allowable, $KL/r < Cc$ (5.899 < 123.583) per AISC E2-1 [Sca]:

$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

$= (1 - (5.9)^2 / (2 \cdot 123.58^2)) 262 /$

$(5/3 + 3 \cdot (5.9) / (8 \cdot 123.58) - (5.9^3) / (8 \cdot 123.58^3))$

= 155.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

$= Sc / Sca + (Rm \cdot C1 / I) / Sba$

$= 1.95 / 155.35 + (20.76 \cdot 137.398 / 22222406) / 174.66$

= 0.020

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.200E+04
Web	278.5	16.0	8.0	44.6	35642.0	762.
Totals	...	...	...	74.6	366522.0	0.276E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 366521.969/74.632

= 49.110 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(49.11, abs(204.0 - 49.11))

= 154.890 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 560.0 - cos(321.53/2)(16.0 + 0.0 + 0.0) - 0.0

= 222.470 mm

Radius of Gyration [r]:

$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$

$= \sqrt{(2763.7 / 74.632)}$

= 60.852 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 222.47/60.852

= 3.656

Unit Force [Force,u]:

$= FI / (2 \cdot \text{Baseplate Length})$

= 426.736 / (2 • 550.0)



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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12,2026

= 0.388 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.388 • 278.453 • 222.47

= 24.032 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 24.032/178427.25

= 1.321 N./mm²

Compressive Allowable, $KL/r < C_c$ (3.6559 < 123.583) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(3.66)²/(2 • 123.58²))262/

(5/3+3•(3.66)/(8• 123.58)-(3.66³)/(8•123.58³)

= 156.1 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 3.3/156.09 + (24.03 • 154.89/2763.653)/174.66

= 0.029

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	2.2586 cm ²
Saddle Load Q0 (Weight)	Q0	2491.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	83.3 Kg
Maximum Transverse Force	Ft	532.9 Kg
Maximum Longitudinal Force	Fl	426.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 426.7/(2.259 • 2)

= 9.3 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 532.9/(2.259 • 2)

= 11.6 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [BlArea_l]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [BlArea_s]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= 426.74/(107.87 • 2.0)

= 0.1940 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:



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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12, 2026

$$\begin{aligned}
 &= B \cdot Ft + \text{Sum of X Moments} \\
 &= 560.0 \cdot 532.89 + 0.0 \\
 &= 298.42 \text{ Kg-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Eccentricity (e):} \\
 &= R_{\text{mom}} / QO \\
 &= 298.42 / 2491.34 \\
 &= 119.78 \text{ mm} > B_{\text{plen}} / 6 \rightarrow \text{Uplift in Transverse direction}
 \end{aligned}$$

$$\begin{aligned}
 f &= B_{\text{plen}} / 2 - \text{Edgedis} \\
 &= 550.0 / 2 - 50.01 \\
 &= 224.99 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Modular Ratio Of Steel/Concrete (n1):} \\
 &= E_S / E_C \\
 &= 203402.5 / 21526.32 \\
 &= 9.45
 \end{aligned}$$

$$\begin{aligned}
 K1 &= 3 (e - 0.5 \cdot B_{\text{plen}}) \\
 &= 3 (119.78 - 0.5 \cdot 550.0) \\
 &= -465.65 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 K2 &= 6 \cdot n1 \cdot A_t / B_{\text{pwid}} \cdot (f + e) \\
 &= 6 \cdot 9.45 \cdot 2.26 / 150.0 \cdot (224.99 + 119.78) \\
 &= 29431.06 \text{ mm}^2
 \end{aligned}$$

$$\begin{aligned}
 K3 &= -K2 \cdot (0.5 \cdot B_{\text{plen}} + f) \\
 &= -29431.06 \cdot (0.5 \cdot 550.0 + 224.99) \\
 &= -14715157.56 \text{ mm}^3
 \end{aligned}$$

$$\begin{aligned}
 &\text{Iteratively Solving for the Effective Bearing Length:} \\
 &Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0 \\
 &Y^3 + -465.65 \cdot Y^2 + 29431.06 \cdot Y + -0.1E+08 = 0 \\
 &Y = 469.69 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Num} &= (B_{\text{plen}} / 2 - Y / 3 - e) \\
 &= (550.0 / 2 - 469.69 / 3 - 119.78) \\
 &= -1.35
 \end{aligned}$$

$$\begin{aligned}
 \text{Denom} &= (B_{\text{plen}} / 2 - Y / 3 + f) \\
 &= (550.0 / 2 - 469.69 / 3 + 224.99) \\
 &= 343.42
 \end{aligned}$$

$$\begin{aligned}
 &\text{Total Bolt Tension Force [Tforce]:} \\
 &= -QO \cdot \text{Num} / \text{Denom} \\
 &= -2491.34 \cdot -1.35 / 343.42 \\
 &= 9.77 \text{ Kg}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\
 &= T_{\text{force}} / (S_{\text{tba}} \cdot N_{\text{bt}}) \\
 &= 9.77 / (137.29 \cdot 1.0) \\
 &= 0.0070 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{Required Area of a Single Bolt [Bltarear]:} \\
 &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\
 &= \max[0.0, 0.194, 0.007] \\
 &= 0.1940 \text{ cm}^2
 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}
 &\text{Bearing Pressure (fc)} \\
 &= 2(QO + T_{\text{force}}) / (Y \cdot B_{\text{pwid}}) \\
 &= 2(2491.34 + 9.77) / (469.69 \cdot 150.0) \\
 &= 696.24 \text{ KPa.}
 \end{aligned}$$



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Saddle Calcs: Operating Case: Step: 18 9:53am Aug 12,2026

Distance from Baseplate Edge to the Web [ADIST]:

$$= (B_{plen} - WebLength) / 2$$

$$= (550.0 - 499.2) / 2$$

$$= 25.4000 \text{ mm}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{force} \cdot A_{dist}$$

$$= 9.77 \cdot 25.4$$

$$= 0.25 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{dist}) / Y$$

$$= 696.24 \cdot (469.69 - 25.4) / 469.69$$

$$= 658.59 \text{ KPa.}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{dist}^2 \cdot B_{pwid} / 6) \cdot (f1 + 2 \cdot f_c)$$

$$= (25.4^2 \cdot 150.0 / 6) \cdot (658.59 + 2 \cdot 696.24)$$

$$= 3.37 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (B_{pwid} \cdot S_{ba}))}$$

$$= \sqrt{(6 \cdot \max(0.25, 3.37) / (150.0 \cdot 207.0))}$$

$$= 2.5285 \text{ mm}$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 9.53, tr = 5.366, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E* / (tg - c) = 0.844, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

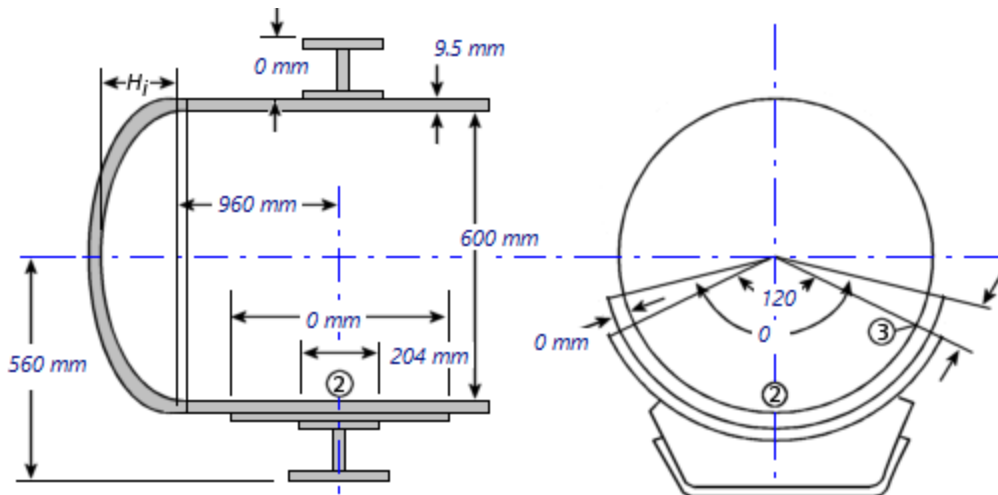
$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{304.765 \cdot 9.53}, 2 \cdot 960.0)$$

$$= 288.0724 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	304.76	mm
Shell Thickness used in this Case	t	9.530	mm
Stiffened Vessel Length per 4.15.6	L	4790.00	mm
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		228.00	N./mm ²
Head Allowable Stress used in Calculation		228.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	560.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		3395.24	Kgf
Pressure used in Analysis	P	3668.193	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	55.70	228.00
Long. Stress at Bottom of Midspan	61.62	228.00
Long. Stress at Top of Saddles	79.18	228.00
Long. Stress at Bottom of Saddles	47.28	228.00



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Saddle Calcs: Test Case:

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Tangential Shear in Shell	8.04	127.92
Circ. Stress at Horn of Saddle	32.10	341.99
Circ. Compressive Stress in Shell	0.92	228.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (351.7/2 + 0) \cdot 560.0/550.0$$

$$= 537.2 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(31.84, 0) \cdot 560.0/2870.0$$

$$= 6.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 2858 + \max(6, 537, 0, 0)$$

$$= 3395.2 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.619 \cdot 1.2)^2)$$

$$= 946.068 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3477.62	Kgf
Transverse Shear Load	175.85	Kgf
Longitudinal Shear Load	31.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 550.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -3395 \cdot 960.0 [1 - (1 - 960.0/4790.0 + (304.765^2 - 0.0^2)/(2 \cdot 960.0 \cdot 4790.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 4790.0))]$$

$$= -620.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 3395 \cdot 4790/4 ((1 + 2(305^2 - 0^2)/(4790^2)) / (1 + (4 \cdot 0)/(3 \cdot 4790)) - 4 \cdot 960/4790)$$

$$= 839.3 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 3668.193 \cdot 304.765/(2 \cdot 9.53) - 839.3/(\pi \cdot 304.8^2 \cdot 9.53)$$

$$= 55.70 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$



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Saddle Calcs: Test Case: Step: 19 9:53am Aug 12,2026

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) + 839.3 / (\pi \cdot 304.8^2 \cdot 9.53)$$

$$= 61.62 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) - 620.3 / (0.1066 \cdot \pi \cdot 304.8^2 \cdot 9.53)$$

$$= 79.18 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) + 620.3 / (0.1923 \cdot \pi \cdot 304.8^2 \cdot 9.53)$$

$$= 47.28 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3395(4790.0 - 2 \cdot 960.0) / (4790.0 + (4 \cdot 0.0 / 3))$$

$$= 2034.3 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2034.31 / (304.765 \cdot 9.53)$$

$$= 8.04 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(304.765 \cdot 9.53)}$$

$$= 42.036 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 3395 \cdot 0.1 / (9.53(204.0 + 42.04 + 42.04))$$

$$= -0.92 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -3395 / (4 \cdot 9.53(204.0 + 42.036 + 42.036)) -$$

$$3 \cdot 0.0529 \cdot 3395 / (2 \cdot 9.53^2)$$

$$= -32.10 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(304.765 \cdot 9.53)}, 2 \cdot 960.0)$$

$$= 288.07 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	550.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(300.0 \cdot 9.53)} + 225.0)^3) / 12$$

$$= 2.224 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$



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Saddle Calcs: Test Case:

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$$= 2.224 + 29.392(97.711 - 4.765)^2$$

$$= 2541.343 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	308.4	9.5	4.8	29.4	14005.2	0.254E+04
Wearplate	225.0	12.0	15.5	27.0	41931.0	0.183E+04
Web	16.0	222.5	132.8	35.6	472579.8	0.191E+04
BasePlate	150.0	16.0	252.0	24.0	604800.1	0.572E+04
Totals	...	...	...	116.0	1133316.0	0.120E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1133316/115.987$$

$$= 97.711 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3395.241$$

$$= 691.0052 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.7826 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 560.0 - 300.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 311.902 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 215.5302 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.7222 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3395 + 82) \cdot 150.0) / (2 \cdot 550.0 \cdot 173.33)}$$

$$= 8.972 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 556.9062 / (3 - 1)$$

$$= 278.453 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 278.4531 \cdot 150.0/2$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12,2026

$$= 208.840 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 3477.625 / (550.0 \cdot 150.0)$$

$$= 4.215 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 208.8 \cdot 4.22$$

$$= 880.323 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 30.08 + 22.276$$

$$= 52.356 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 880.3 / 52.3563$$

$$= 1.649 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.145E+04
Web	139.2	16.0	8.0	22.3	17821.0	770.
Totals	...	...	...	52.4	348700.9	0.222E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 204.0 - 16.0$$

$$= 188.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 348700.938 / 52.356$$

$$= 66.602 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(66.602, 204.0 - 66.602)$$

$$= 137.398 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(2222.2 / 52.356)}$$

$$= 65.150 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 560.0 - \cos(120.0/2) (300.0 + 9.53 + 12.0) - 16.0$$

$$= 383.235 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 550.0 - 50.8$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12,2026

= 499.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (556.906 - 499.2)/2

= 28.853 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$

= $\sqrt{(383.235^2 + 28.853^2)}$

= 384.320 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \cdot \pi^2 \cdot 0.20270\text{E}+09/260.0)}$

= 124.057

Slenderness ratio [KL/r]:

= KL/r

= 1 • 384.32/65.15

= 5.899

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 31.8 / (2 • 550.0) • 278.453 • 384.32/2

= 1.549 Kg-m.

Compressive Allowable, KL/r < Cc (5.899 < 124.0574) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (5.9)² / (2 • 124.06²)) 260 /

(5/3 + 3 • (5.9) / (8 • 124.06) - (5.9³) / (8 • 124.06³))

= 154.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 1.65/154.17 + (1.55 • 137.398/22222406) / 173.33

= 0.011

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.200E+04
Web	278.5	16.0	8.0	44.6	35642.0	762.
Totals	...	...	...	74.6	366522.0	0.276E+04

Distance to Centroid from Datum [ytot]:

= AY / A

= 366521.969/74.632

= 49.110 mm

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))

= max(49.11, abs(204.0 - 49.11))

= 154.890 mm

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk

= 560.0 - cos(321.53/2) (16.0 + 0.0 + 0.0) - 0.0

= 222.470 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{(2763.7/74.632)}$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12, 2026

= 60.852 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 222.47/60.852

= 3.656

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 31.836/(2 • 550.0)

= 0.029 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 0.029 • 278.453 • 222.47

= 1.793 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1.793/178427.25

= 0.099 N./mm²

Compressive Allowable, KL/r < Cc (3.6559 < 124.0574) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))

= (1-(3.66)²/(2 • 124.06²))260/

(5/3+3•(3.66)/(8• 124.06)-(3.66³)/(8•124.06³))

= 154.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba

= 2.79/154.9 + (1.79 • 154.89/2763.653)/173.33

= 0.019

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	2.2586 cm ²
Saddle Load QO (Weight)	QO	2940.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	6.2 Kg
Maximum Transverse Force	Ft	175.9 Kg
Maximum Longitudinal Force	F1	31.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)

= 31.8/(2.259 • 2)

= 0.7 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)

= 175.9/(2.259 • 2)

= 3.8 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (QO > QL --> No Uplift in Longitudinal direction)



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Saddle Calcs: Test Case:

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Bolt Area due to Shear Load [Bltarears]:

= $F_l / (BoltShearAllowable \cdot Nbolts)$

= $31.84 / (103.2 \cdot 2.0)$

= 0.0151 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot Ft + \text{Sum of X Moments}$

= $560.0 \cdot 175.85 + 0.0$

= 98.48 Kg-m.

Eccentricity (e):

= $Rmom / QO$

= $98.48 / 2940.47$

= 33.49 mm < $B_{plen}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[Bltarearl, Bltarears, Bltareart]$

= $\max[0.0, 0.0151, 0.0]$

= 0.0151 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$

= $\min(204.0 + 1.56 \cdot \sqrt{304.765 \cdot 9.53}, 2 \cdot 960.0)$

= 288.0724 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	304.76	mm
Shell Thickness used in this Case	t	9.530	mm
Stiffened Vessel Length per 4.15.6	L	4790.00	mm
Distance from Saddle to Vessel tangent	a1 or a	960.00	mm
Saddle Width	b1 or b	204.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		228.00	N./mm ²
Head Allowable Stress used in Calculation		228.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	560.00	mm
Coefficient of Friction	mu	0.50	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		3381.95	Kgf
Pressure used in Analysis	P	3668.193	KPa.

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	55.71	228.00



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Saddle Calcs: Test Case: Step: 19 9:53am Aug 12,2026

Long. Stress at Bottom of Midspan	61.61	228.00
Long. Stress at Top of Saddles	79.10	228.00
Long. Stress at Bottom of Saddles	47.33	228.00
Tangential Shear in Shell	8.01	127.92
Circ. Stress at Horn of Saddle	31.97	341.99
Circ. Compressive Stress in Shell	0.92	228.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(351.7/2 + 0) \cdot 560.0/550.0$$

$$= 537.2 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(31.84, 0) \cdot 560.0/2870.0$$

$$= 6.2 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 2845 + \max(6, 537, 0, 0)$$

$$= 3382.0 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.619 \cdot 1.2)^2)$$

$$= 946.068 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 3464.34 Kgf

Transverse Shear Load 175.85 Kgf

Longitudinal Shear Load 31.84 Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 550.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1066 \quad K2 = 1.1707 \quad K3 = 0.8799 \quad K4 = 0.4011$$

$$K5 = 0.7603 \quad K6 = 0.0529 \quad K7 = 0.0529 \quad K8 = 0.3405$$

$$K9 = 0.2711 \quad K10 = 0.0581 \quad K1^* = 0.1923$$

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -3382 \cdot 960.0 [1 - (1 - 960.0/4790.0 + (304.765^2 - 0.0^2) /$$

$$(2 \cdot 960.0 \cdot 4790.0) / (1 + (4 \cdot 0.0) / (3 \cdot 4790.0))]$$

$$= -617.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 3382 \cdot 4790 / 4 ((1 + 2(305^2 - 0^2)/(4790^2)) / (1 + (4 \cdot 0) /$$

$$(3 \cdot 4790)) - 4 \cdot 960 / 4790)$$

$$= 836.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 t)$$

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) - 836.0 / (\pi \cdot 304.8^2 \cdot 9.53)$$

$$= 55.71 \text{ N./mm}^2$$



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Saddle Calcs: Test Case:

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Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) + 836.0 / (\pi \cdot 304.8^2 \cdot 9.53)$$

$$= 61.61 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) - 617.9 / (0.1066 \cdot \pi \cdot 304.8^2 \cdot 9.53)$$

$$= 79.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 3668.193 \cdot 304.765 / (2 \cdot 9.53) + 617.9 / (0.1923 \cdot \pi \cdot 304.8^2 \cdot 9.53)$$

$$= 47.33 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3382(4790.0 - 2 \cdot 960.0) / (4790.0 + (4 \cdot 0.0 / 3))$$

$$= 2026.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2026.35 / (304.765 \cdot 9.53)$$

$$= 8.01 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(304.765 \cdot 9.53)}$$

$$= 42.036 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 3382 \cdot 0.1 / (9.53(204.0 + 42.04 + 42.04))$$

$$= -0.92 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -3382 / (4 \cdot 9.53(204.0 + 42.036 + 42.036)) - 3 \cdot 0.0529 \cdot 3382 / (2 \cdot 9.53^2)$$

$$= -31.97 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(204.0 + 1.56 \cdot \sqrt{(304.765 \cdot 9.53)}, 2 \cdot 960.0)$$

$$= 288.07 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	550.0000	mm
Baseplate Thickness	Bpthk	16.0000	mm
Baseplate Width	Bpwid	150.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm
Friction Coefficient	mu	0.500	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(300.0 \cdot 9.53)} + 225.0)9.53^3) / 12$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12,2026

$$= 2.224 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2.224 + 29.392(97.711 - 4.765)^2$$

$$= 2541.343 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	308.4	9.5	4.8	29.4	14005.2	0.254E+04
Wearplate	225.0	12.0	15.5	27.0	41931.0	0.183E+04
Web	16.0	222.5	132.8	35.6	472579.8	0.191E+04
BasePlate	150.0	16.0	252.0	24.0	604800.1	0.572E+04
Totals	...	...	...	116.0	1133316.0	0.120E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1133316/115.987$$

$$= 97.711 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 3381.955$$

$$= 688.3011 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.7795 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 560.0 - 300.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 311.902 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 214.6868 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 1.7155 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(3382 + 82)150.0) / (2 \cdot 550.0 \cdot 173.33)}$$

$$= 8.955 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 556.9062 / (3 - 1)$$

$$= 278.453 \text{ mm}$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12,2026

Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 278.4531 \cdot 150.0 / 2$$

$$= 208.840 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 3464.338 / (550.0 \cdot 150.0)$$

$$= 4.199 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 208.8 \cdot 4.2$$

$$= 876.960 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 30.08 + 22.276$$

$$= 52.356 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 877.0 / 52.3563$$

$$= 1.643 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.145E+04
Web	139.2	16.0	8.0	22.3	17821.0	770.
Totals	...	...	...	52.4	348700.9	0.222E+04

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 204.0 - 16.0$$

$$= 188.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 348700.938 / 52.356$$

$$= 66.602 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(ytot, Saddle \text{ Width} - ytot)$$

$$= \max(66.602, 204.0 - 66.602)$$

$$= 137.398 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (2222.2 / 52.356) }$$

$$= 65.150 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 560.0 - \cos(120.0 / 2) (300.0 + 9.53 + 12.0) - 16.0$$

$$= 383.235 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(120.0 / 2) (300.0 + 9.53 + 12.0)$$

$$= 556.906 \text{ mm}$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12, 2026

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 550.0 - 50.8$$

$$= 499.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (556.906 - 499.2) / 2$$

$$= 28.853 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(383.235^2 + 28.853^2)}$$

$$= 384.320 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.20270\text{E}+09 / 260.0)}$$

$$= 124.057$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 384.32 / 65.15$$

$$= 5.899$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 31.8 / (2 \cdot 550.0) \cdot 278.453 \cdot 384.32 / 2$$

$$= 1.549 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($5.899 < 124.0574$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.9)^2 / (2 \cdot 124.06^2)) 260 /$$

$$(5/3 + 3 \cdot (5.9) / (8 \cdot 124.06) - (5.9^3) / (8 \cdot 124.06^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.64 / 154.17 + (1.55 \cdot 137.398 / 22222406) / 173.33$$

$$= 0.011$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	188.0	110.0	30.1	330880.0	0.200E+04
Web	278.5	16.0	8.0	44.6	35642.0	762.
Totals	...	...	...	74.6	366522.0	0.276E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 366521.969 / 74.632$$

$$= 49.110 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(49.11, \text{abs}(204.0 - 49.11))$$

$$= 154.890 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 560.0 - \cos(321.53/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 222.470 \text{ mm}$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12,2026

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{2763.7/74.632}$$

$$= 60.852 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 222.47/60.852$$

$$= 3.656$$

Unit Force [Force,u]:

$$= F_l / (2 \cdot \text{Baseplate Length})$$

$$= 31.836 / (2 \cdot 550.0)$$

$$= 0.029 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.029 \cdot 278.453 \cdot 222.47$$

$$= 1.793 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1.793/178427.25$$

$$= 0.099 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($3.6559 < 124.0574$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (3.66)^2 / (2 \cdot 124.06^2)) 260 /$$

$$(5/3 + 3 \cdot (3.66) / (8 \cdot 124.06) - (3.66)^3 / (8 \cdot 124.06^3))$$

$$= 154.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 2.78/154.9 + (1.79 \cdot 154.89/2763.653) / 173.33$$

$$= 0.018$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.0 mm
Nominal Bolt Diameter	Bnd	24.0000 mm
Thread Series	Series	TEMA Metric
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	2.2586 cm ²
Saddle Load Q0 (Weight)	Q0	2927.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	6.2 Kg
Maximum Transverse Force	Ft	175.9 Kg
Maximum Longitudinal Force	F1	31.8 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 31.8 / (2.259 \cdot 2)$$

$$= 0.7 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 175.9 / (2.259 \cdot 2)$$

$$= 3.8 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$



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Saddle Calcs: Test Case:

Step: 19 9:53am Aug 12, 2026

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $31.84 / (103.2 \cdot 2.0)$
= 0.0151 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $560.0 \cdot 175.85 + 0.0$
= 98.48 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $98.48 / 2927.19$
= 33.64 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 0.0151, 0.0]$
= 0.0151 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1- Channel inl

Nozl: 9 9:53am Aug 12,2026

Input, Nozzle Desc: N1- Channel inlet

From: 20

Pressure for Reinforcement Calculations	P	2005.881	KPa.
Temperature for Internal Pressure	Temp	66	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	9.5300	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		376.20	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.5300	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	308.2750	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



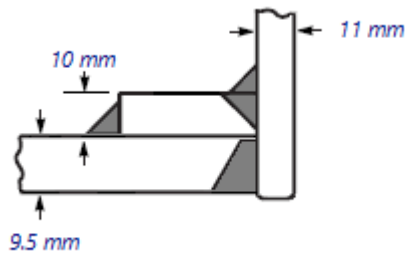
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Nozzle Calcs.: N1- Channel inl

Nozl: 9 9:53am Aug 12,2026



Reinforcement CALCULATION, Description: N1- Channel inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2005.88 \cdot 303.175 / (118 \cdot 1.0 - 0.6 \cdot 2005.88)$
 $= 5.2072 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2005.88 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 2005.88)$
 $= 1.4207 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm
Parallel to Vessel Wall, opening length	d	152.6794 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.8875 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

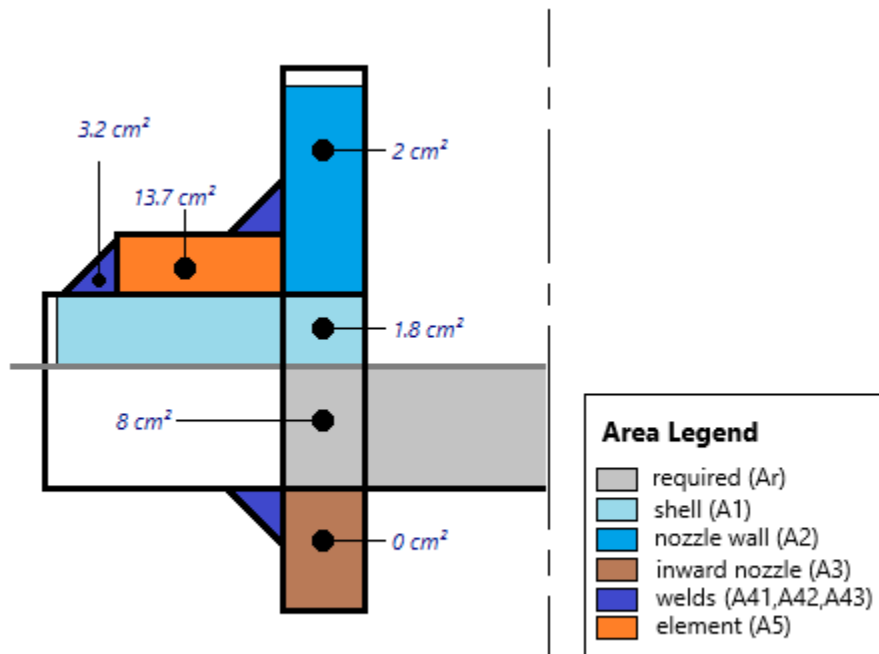


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.950	NA	NA
Area in Shell	A1	1.752	NA	NA
Area in Nozzle Wall	A2	2.026	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	3.186	NA	NA
Area in Element	A5	13.708	NA	NA
TOTAL AREA AVAILABLE	Atot	20.673	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (152.6794 \cdot 5.2072 \cdot 1.0 + 2 \cdot 7.7978 \cdot 5.2072 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 7.950 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 152.679(1.0 \cdot 6.355 - 1.0 \cdot 5.207) - 2 \cdot 7.798 \\
 &\quad (1.0 \cdot 6.355 - 1.0 \cdot 5.207) \cdot (1 - 1.0) \\
 &= 1.752 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 15.89) \cdot (7.8 - 1.42) \cdot 1.0
 \end{aligned}$$



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 Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

= 2.026 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
 = (3.1859) • 1.0 + (0.0) • 1.0 + 0.0² • 1.0
 = 3.186 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (305.3588 - 168.275) • 10.0 • 1.0
 = 13.708 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.5957 mm
Wall Thickness per UG16(b),	tr16b = 5.6750 mm
Wall Thickness, shell/head, internal pressure	trb1 = 8.3822 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 8.3822 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.6750 mm
Wall Thickness per table UG-45	tb3 = 9.3950 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[9.395, max(8.3822, 5.675)]
 = 8.3822 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.5957, 8.3822)
 = 8.3822 mm

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	8.4,	Allowable :	118.0 N./mm ²	Passed
Expansion :	0.0,	Allowable :	286.6 N./mm ²	Passed
Occasional :	9.3,	Allowable :	156.9 N./mm ²	Passed
Shear :	4.6,	Allowable :	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 1.421, c = 3.175 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.221, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 1.421, c = 3.175 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.221, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 1.421, c = 3.175 mm, E* = 1.0



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

Thickness Ratio = $tr * E^* / (tg - c) = 0.221$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 9.53$, $tr = 5.207$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (tg - c) = 0.819$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 9.53$, $tr = 5.207$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (tg - c) = 0.819$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

Gov. MDMT of the Nozzle : -39 °C
Gov. MDMT of the Reinforcement Pad : -39 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $2005.88 / 5100.00 = 0.393$

Weld Size Calculations, Description: N1- Channel inl

Intermediate Calc. for nozzle/shell Welds T_{min} 7.7978 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 6.3550 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 * t_{min}$	$21.2100 = 0.7 * W_o$ mm
Pad Weld	$3.1775 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (7.9503 - 1.7525 + 2 \cdot 7.7978 \cdot 1.0 \cdot (1.0 \cdot 6.355 - 5.2072)) \cdot 118)$
 $= 7672.71$ Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2.0263 + 13.7084 + 3.1859 - 0.0 \cdot 1.0) \cdot 118$
 $= 22765.64$ Kgf

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.0263 + 0.0 + 7.0084 + (0.9911)) \cdot 118$
 $= 12063.25$ Kgf

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

$$= (2.0263 + 0.0 + 3.1859 + 13.7084 + (0.9911)) \cdot 118$$

$$= 23958.15 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 30.0 \cdot 0.49 \cdot 118$$

$$= 46752. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 308.275 \cdot 8.0 \cdot 0.49 \cdot 118$$

$$= 22840. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 80.2386) \cdot (10.9728 - 3.175) \cdot 0.7 \cdot 118$$

$$= 16556. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 168.275 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 27524. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (9.53 - 3.175) \cdot 0.74 \cdot 118$$

$$= 14957. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (22840 + 16556) = 39395 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (46752 + 27524 + 14957 + 0) = 89233 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (22840 + 14957 + 0) = 37796 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 39395	Kgf, must exceed W = 7672 Kgf	or W1 = 22765 Kgf
Path 2-2 = 89232	Kgf, must exceed W = 7672 Kgf	or W2 = 12063 Kgf
Path 3-3 = 37796	Kgf, must exceed W = 7672 Kgf	or W3 = 23958 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2437 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.0402 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.5701 mm

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.1750 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	600.000 mm
Shell Thickness	9.5300 mm



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

Shell Stress Concentration Factor		1.000	
Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.1750	mm
Nozzle Diameter		168.275	mm
Nozzle Thickness		10.9728	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	308.275	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2005.88	KPa.
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	0.28	Kg-m.
Longitudinal Moment	Ml	0.37	Kg-m.
Torsional Moment	Mt	0.46	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(311.353(9.53 - 3.175))}$$

$$= 315.065 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	622.705	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	16.355	mm
Nozzle Thickness used	(t) =	7.798	mm
T / t	=	2.097	
d / t	=	21.580	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.667$$

Nr/P	=	0.077
Mr/P	=	0.087
M0/P	=	0.053
N0/P	=	0.154
MrD/Mc	=	0.196
NrDL/Mc	=	0.131
M0d/Mc	=	0.142



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

NØDL/Mc = 0.151
MrD/Ml = 0.134
NrDL/Ml = 0.085
MØD/Ml = 0.093
NØDL/Ml = 0.266

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	36	38	36	38
Outplane Stress Summary	43	35	43	35

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	26	11	27	11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	43	35	43	35

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

Normal Pressure Stress	18	18	18	18
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Outplane Stress Summary	25	15	25	15
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	38	36	38

Inplane Stress Summary	45	31	45	31
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	45	31	45	31
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24

Hoop Stress Summary	24	27	24	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

Axial Stress Summary	31	-6	31	-7
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	2	2	-2	-2
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	32	34	32	34
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24

Hoop Stress Summary	24	27	24	27
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	31	-6	31	-7
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-2	-2	2	2
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	32	34	32	34
------------------------	----	----	----	----



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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{306.352(9.53 - 3.175)}$$

$$= 313.882 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	612.705 mm
Nozzle Outside Diameter	(d) =	308.275 mm
Vessel Thickness used	(T) =	6.355 mm
Nozzle Thickness used	(t) =	77.798 mm
T / t	=	0.082
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.940$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.071
M0/P	=	0.020
N0/P	=	0.027
MrD/Mc	=	0.241
NrDL/Mc	=	0.099
M0d/Mc	=	0.072
N0DL/Mc	=	0.053
MrD/M1	=	0.050
NrDL/M1	=	0.048
M0D/M1	=	0.015
N0DL/M1	=	0.027

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	94	96	94	96
Outplane Stress Summary	109	87	109	87

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	41	-41	41	-41
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	93	10	93	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N1- Channel inl Nozl: 9 9:53am Aug 12,2026

(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			109	87	109	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			2	2	2	2
Outplane Bending (P)			11	-11	11	-11
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			0	0	0	0
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			47	47	47	47
Outplane Stress Summary			61	38	61	38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			41	-41	41	-41
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			0	0	0	0
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			94	96	94	96
Inplane Stress Summary			140	59	141	59

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			140	59	141	59



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	38	36	38	36	38	36	38
Circ. Pl (SUS)	2	2	2	2	1	1	1	1
Circ. Q (SUS)	4	-4	4	-4	7	-7	7	-7
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	1	1	1	1	2	2	2	2
Long. Q (SUS)	7	-7	7	-7	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	36.0	38.0	36.0	38.0	36.0	38.0	36.0	38.0
Pm+Pl (SUS)	38.1	40.0	38.1	40.0	37.1	39.1	37.1	39.1
Pm+Pl+Q (Total)	42.1	36.0	42.1	36.0	44.0	32.1	44.0	32.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	118.00	Passed
Pm+Pl (SUS)	40.05	177.00	Passed
Pm+Pl+Q (TOTAL)	44.05	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	94	96	94	96	94	96	94	96
Circ. Pl (SUS)	2	2	2	2	4	4	4	4
Circ. Q (SUS)	11	-11	11	-11	41	-41	41	-41
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	4	4	4	4	2	2	2	2
Long. Q (SUS)	41	-41	41	-41	11	-11	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	94.0	96.0	94.0	96.0	94.0	96.0	94.0	96.0
Pm+Pl (SUS)	96.0	98.0	96.0	98.0	98.0	100.0	98.0	100.0
Pm+Pl+Q (Total)	107.1	87.0	107.1	87.0	139.0	59.0	139.0	59.0

Vessel Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N1- Channel in1 Nozl: 9 9:53am Aug 12,2026

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	96.00	118.00	Passed
Pm+P1 (SUS)	100.02	177.00	Passed
Pm+P1+Q (TOTAL)	139.01	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	24	22	24	22	24	22	24
Circ. P1 (SUS)	2	2	2	2	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. P1 (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	19	-19	19	-19	19	-19	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	22.0	24.0	22.0	24.0	22.0	24.0	22.0	24.0
Pm+P1 (SUS)	24.3	26.3	24.3	26.3	24.3	26.3	24.3	26.3
Pm+P1+Q (Total)	31.5	33.2	31.5	33.2	31.5	33.2	31.5	33.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	118.00	Passed
Pm+P1 (SUS)	26.28	177.00	Passed
Pm+P1+Q (TOTAL)	33.24	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N1- Shell outle

Nozl: 10

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Input, Nozzle Desc: N1- Shell outlet

From: 40

Pressure for Reinforcement Calculations	P	2600.000	KPa.
Temperature for Internal Pressure	Temp	166	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	140.75	N./mm ²
Parent Allowable Stress At Ambient	Sva	161.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	9.5300	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1320.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.5300	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	413.0500	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



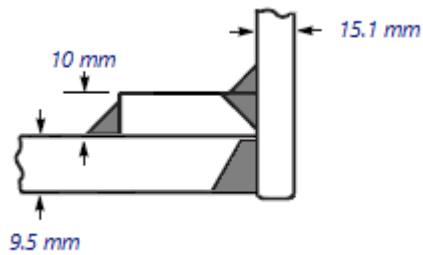
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Nozzle Calcs.: N1- Shell outlet

Nozl: 10 9:53am Aug 12,2026



Reinforcement CALCULATION, Description: N1- Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2600.0 \cdot 303.175 / (141 \cdot 1.0 - 0.6 \cdot 2600.0)$
 $= 5.6633 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2600.0 \cdot 136.525 / (118 \cdot 1.0 + 0.4 \cdot 2600.0)$
 $= 2.9821 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	498.4496 mm
Parallel to Vessel Wall, opening length	d	249.2248 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.8875 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 140.8)$
 $= 0.838$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 140.8)$
 $= 0.838$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 140.8)$
 $= 0.980$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.838, 0.98)$
 $= 0.838$



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Nozzle Calcs.: N1- Shell outle

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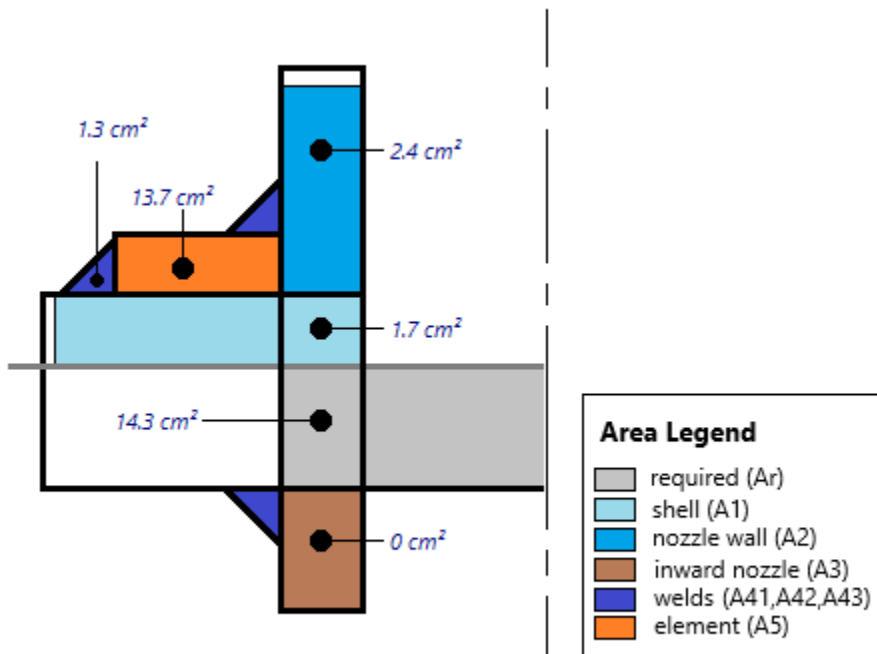


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	14.333	NA	NA
Area in Shell	A1	1.697	NA	NA
Area in Nozzle Wall	A2	2.379	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		1.324	NA	NA
Area in Element	A5	13.726	NA	NA
TOTAL AREA AVAILABLE	Atot	19.126	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 364.1594 10.0000 mm

Based on given Pad Diameter: 413.0500 6.5078 mm

Based on Shell or Nozzle Thickness: 368.6527 9.5300 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (249.2248 \cdot 5.6633 \cdot 1.0 + 2 \cdot 11.9126 \cdot 5.6633 \cdot 1.0 \cdot (1 - 0.84))$$

$$= 14.333 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 249.225(1.0 \cdot 6.355 - 1.0 \cdot 5.663) - 2 \cdot 11.913$$

$$(1.0 \cdot 6.355 - 1.0 \cdot 5.663) \cdot (1 - 0.838)$$



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

$$= 1.697 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 15.89) \cdot (11.91 - 2.98) \cdot 0.8383$$

$$= 2.379 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4$$

$$= (0.8309) \cdot 0.84 + (0.0) \cdot 0.84 + 8.0^2 \cdot 0.98$$

$$= 1.324 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4$$

$$= (413.05 - 273.05) \cdot 10.0 \cdot 0.9804$$

$$= 13.726 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $ta = 6.1571 \text{ mm}$

Wall Thickness per UG16(b), $tr16b = 5.6750 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb1 = 8.8383 \text{ mm}$

Wall Thickness $tb1 = \max(trb1, tr16b) = 8.8383 \text{ mm}$

Wall Thickness $tb2 = \max(trb2, tr16b) = 5.6750 \text{ mm}$

Wall Thickness per table UG-45 $tb3 = 11.2776 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[11.278, \max(8.8383, 5.675)]$$

$$= 8.8383 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(6.1571, 8.8383)$$

$$= 8.8383 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	282.7 N./mm ²	Passed
Occasional	:	13.0,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	3.0,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $tg = 13.202$, $tr = 2.982$, $c = 3.175 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.297$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $0 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $tg = 10.0$, $c = 3.175 \text{ mm}$, $E* = 1.0$

Thickness Ratio = $tr \cdot E* / (tg - c) = 0.778$, Temp. Reduction = $12 \text{ }^\circ\text{C}$

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B $-29 \text{ }^\circ\text{C}$



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.53, tr = 4.944, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.53, tr = 4.944, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Gov. MDMT of the Nozzle : -41 °C

Gov. MDMT of the Reinforcement Pad : -41 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2600.00 / 5100.00 = 0.510$

Weld Size Calculations, Description: N1- Shell outle

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm

Intermediate Calc. for pad/shell Welds TminPad 6.3550 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$3.1775 = 0.5 * \text{TminPad}$	$5.6560 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (14.3326 - 1.6972 + 2 \cdot 11.9126 \cdot 0.8383 \cdot (1.0 \cdot 6.355 - 5.6633)) \cdot 141)$
 $= 18333.43 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (2.3789 + 13.7258 + 1.324 - 0.0 \cdot 0.84) \cdot 141$
 $= 25014.83 \text{ Kgf}$

Weld Load [W2]:



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (2.3789 + 0.0 + 0.8383 + (1.2693)) \cdot 141$$

$$= 6439.34 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (2.3789 + 0.0 + 1.324 + 13.7258 + (1.2693)) \cdot 141$$

$$= 26836.62 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 25287. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 413.05 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 35789. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 130.5687) \cdot (15.0876 - 3.175) \cdot 0.7 \cdot 118$$

$$= 41157. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 273.05 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 44662. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 273.05 \cdot (9.53 - 3.175) \cdot 0.74 \cdot 141$$

$$= 28949. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (35789 + 41157) = 76946 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (25287 + 44662 + 28949 + 0) = 98899 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (35789 + 28949 + 0) = 64739 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 76945	Kgf, must exceed W = 18333 Kgf	or W1 = 25014 Kgf
Path 2-2 = 98898	Kgf, must exceed W = 18333 Kgf	or W2 = 6439 Kgf
Path 3-3 = 64738	Kgf, must exceed W = 18333 Kgf	or W3 = 26836 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2913 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 32.8654 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 292.3954 mm

Input Echo, WRC297 Item 1, Description: N1- Shell outl:



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.1750	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	600.000	mm
Shell Thickness	9.5300	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-106 B	
Vessel Cold S.I. Allowable	Smc 161.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 140.75	N./mm ²

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.1750	mm
Nozzle Diameter	273.050	mm
Nozzle Thickness	15.0876	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	413.050	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2600.00	KPa.
Radial Load	P	-683.25	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	0.76	Kg-m.
Longitudinal Moment	Ml	1.02	Kg-m.
Torsional Moment	Mt	1.27	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(311.353(9.53 - 3.175))}$$

$$= 419.840 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	622.705	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	16.355	mm
Nozzle Thickness used	(t) =	11.913	mm
T / t	=	1.373	
d / t	=	22.921	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.706$$

Nr/P	=	0.058
Mr/P	=	0.092
M0/P	=	0.037
N0/P	=	0.126



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

MrD/Mc = 0.222
 NrDL/Mc = 0.104
 MØd/Mc = 0.102
 NØDL/Mc = 0.130
 MrD/Ml = 0.108
 NrDL/Ml = 0.054
 MØD/Ml = 0.050
 NØDL/Ml = 0.190

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	46	49	46	49
Outplane Stress Summary	55	47	55	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	38	11	38	10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	55	47	55	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	32	21	32	20
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	13	-13	13	-13
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	49	46	49

Inplane Stress Summary	62	37	62	36
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
----------------------	----	----	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	62	37	62	37
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	34	31	34

Hoop Stress Summary	34	37	34	37
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	34	-1	34	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	36	39	36	39

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	31	34	31	34
Hoop Stress Summary	34	37	34	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	34	-1	34	-1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



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Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

Two * Max Shear Stress 36 39 36 39

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{306.352(9.53 - 3.175)}$$

$$= 418.657 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 612.705 mm
 Nozzle Outside Diameter (d) = 413.050 mm
 Vessel Thickness used (T) = 6.355 mm
 Nozzle Thickness used (t) = 81.913 mm
 T / t = 0.078
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 6.619$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.027
 MrD/Mc = 0.239
 NrDL/Mc = 0.092
 M0d/Mc = 0.067
 N0DL/Mc = 0.049
 MrD/Ml = 0.039
 NrDL/Ml = 0.036
 M0D/Ml = 0.012
 N0DL/Ml = 0.022

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	19	-19	19	-19
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	122	125	122	125
Outplane Stress Summary	146	110	146	110

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	138	-1	139	-1



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Nozzle Calcs.: N1- Shell outle

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			2	2	-2	-2
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			147	112	147	112

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			4	4	4	4
Outplane Bending (P)			19	-19	19	-19
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			0	0	0	0
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			61	61	61	61
Outplane Stress Summary			84	46	85	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			7	7	7	7
Inplane Bending (P)			70	-70	70	-70
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			0	0	0	0
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			122	125	122	125
Inplane Stress Summary			199	63	201	61

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-2	-2	2	2
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-2	-2	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside



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 Nozzle Calcs.: N1- Shell outle Nozl: 10 9:53am Aug 12,2026

Two * Max Shear Stress 199 63 201 62

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	49	46	49	46	49	46	49
Circ. Pl (SUS)	3	3	3	3	1	1	1	1
Circ. Q (SUS)	5	-5	5	-5	13	-13	13	-13
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	1	1	1	1	3	3	3	3
Long. Q (SUS)	13	-13	13	-13	5	-5	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	46.0	49.0	46.0	49.0	46.0	49.0	46.0	49.0
Pm+Pl (SUS)	49.0	52.0	49.0	52.0	47.0	50.0	47.0	50.0
Pm+Pl+Q (Total)	54.1	47.0	54.1	47.0	60.0	37.1	60.0	37.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	140.75	Passed
Pm+Pl (SUS)	52.04	211.13	Passed
Pm+Pl+Q (TOTAL)	60.03	422.26	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122	125	122	125	122	125	122	125
Circ. Pl (SUS)	4	4	4	4	7	7	7	7
Circ. Q (SUS)	19	-19	19	-19	70	-70	70	-70
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	7	7	7	7	4	4	4	4
Long. Q (SUS)	70	-70	70	-70	19	-19	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	122.0	125.0	122.0	125.0	122.0	125.0	122.0	125.0
Pm+Pl (SUS)	126.1	129.1	126.1	129.1	129.1	132.1	129.1	132.1
Pm+Pl+Q (Total)	145.5	112.1	145.5	112.1	199.0	62.2	199.0	62.2



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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	140.75	Passed
Pm+P1 (SUS)	132.06	211.13	Passed
Pm+P1+Q (TOTAL)	199.03	422.26	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	34	31	34	31	34	31	34
Circ. P1 (SUS)	3	3	3	3	3	3	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. P1 (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	17	-17	17	-17	17	-17	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	31.0	34.0	31.0	34.0	31.0	34.0	31.0	34.0
Pm+P1 (SUS)	34.1	37.0	34.1	37.0	34.1	37.0	34.1	37.0
Pm+P1+Q (Total)	34.4	39.1	34.4	39.1	34.4	39.1	34.4	39.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	34.00	118.00	Passed
Pm+P1 (SUS)	37.05	177.00	Passed
Pm+P1+Q (TOTAL)	39.05	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2- shell inlet

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Input, Nozzle Desc: N2- shell inlet

From: 40

Pressure for Reinforcement Calculations	P	2600.000	KPa.
Temperature for Internal Pressure	Temp	166	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	140.75	N./mm ²
Parent Allowable Stress At Ambient	Sva	161.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	9.5300	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		4820.00	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		12.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.5300	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	463.8500	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



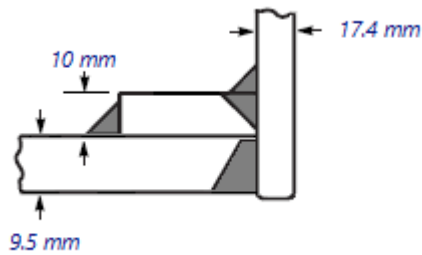
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Nozzle Calcs.: N2- shell inlet

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Reinforcement CALCULATION, Description: N2- shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	12.750 in.
Actual Thickness Used in Calculation	0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2600.0 \cdot 303.175 / (141 \cdot 1.0 - 0.6 \cdot 2600.0)$
 $= 5.6633 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2600.0 \cdot 161.925 / (118 \cdot 1.0 + 0.4 \cdot 2600.0)$
 $= 3.5369 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	590.6008 mm
Parallel to Vessel Wall, opening length	d	295.3004 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		15.8875 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 140.8)$
 $= 0.838$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 140.8)$
 $= 0.838$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 140.8)$
 $= 0.980$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.838, 0.98)$
 $= 0.838$



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Nozzle Calcs.: N2- shell inlet

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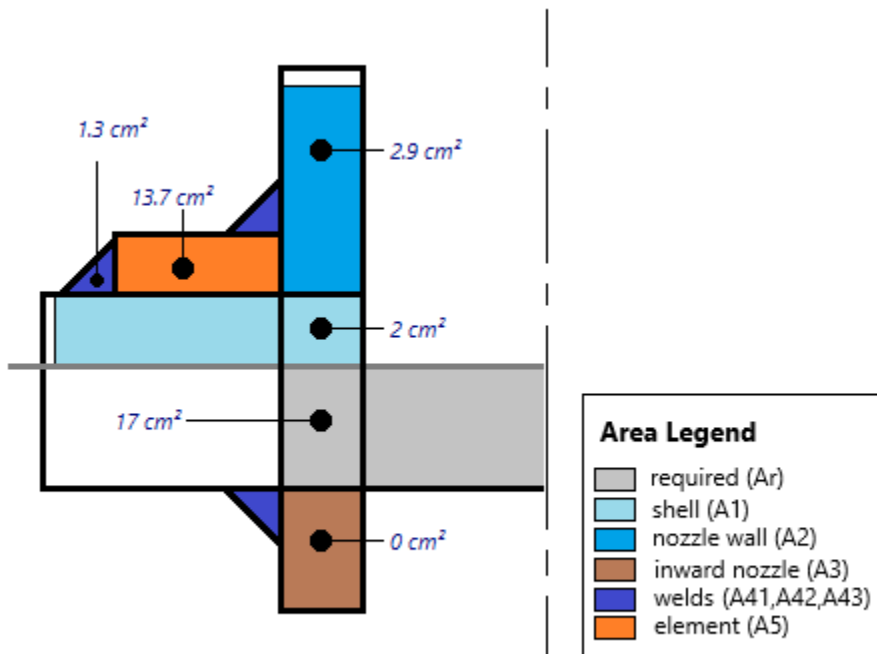


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.985	NA	NA
Area in Shell	A1	2.011	NA	NA
Area in Nozzle Wall	A2	2.860	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.324	NA	NA
Area in Element	A5	13.726	NA	NA
TOTAL AREA AVAILABLE	Atot	19.921	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness

Based on given Pad Thickness: 433.9086 10.0000 mm

Based on given Pad Diameter: 463.8501 7.8613 mm

Based on Shell or Nozzle Thickness: 439.3365 9.5300 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (295.3004 \cdot 5.6633 \cdot 1.0 + 2 \cdot 14.2748 \cdot 5.6633 \cdot 1.0 \cdot (1 - 0.84))$$

$$= 16.985 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 295.3(1.0 \cdot 6.355 - 1.0 \cdot 5.663) - 2 \cdot 14.275$$

$$(1.0 \cdot 6.355 - 1.0 \cdot 5.663) \cdot (1 - 0.838)$$



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Nozzle Calcs.: N2- shell inlet

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$$= 2.011 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 15.89) \cdot (14.27 - 3.54) \cdot 0.8383$$

$$= 2.860 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (W_o^2 - A_r \text{ Lost}) \cdot f_{r3} + ((W_i - c_{an}/0.707)^2 - A_r \text{ Lost}) \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= (0.8309) \cdot 0.84 + (0.0) \cdot 0.84 + 8.0^2 \cdot 0.98$$

$$= 1.324 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (463.8501 - 323.85) \cdot 10.0 \cdot 0.9804$$

$$= 13.726 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 6.7119 \text{ mm}$

Wall Thickness per UG16(b), $t_{r16b} = 5.6750 \text{ mm}$

Wall Thickness, shell/head, internal pressure $t_{rb1} = 8.8383 \text{ mm}$

Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 8.8383 \text{ mm}$

Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.6750 \text{ mm}$

Wall Thickness per table UG-45 $t_{b3} = 11.5062 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[11.506, \max(8.8383, 5.675)]$$

$$= 8.8383 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(6.7119, 8.8383)$$

$$= 8.8383 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	12.3,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	282.7 N./mm ²	Passed
Occasional	:	12.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	2.6,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 15.269$, $t_r = 3.537$, $c = 3.175 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.292$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 4°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 10.0$, $c = 3.175 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.778$, Temp. Reduction = 12°C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29°C



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Nozzle Calcs.: N2- shell inlet Nozl: 11 9:53am Aug 12,2026

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 9.53, tr = 4.944, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.53, tr = 4.944, c = 3.175 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.778$, Temp. Reduction = 12 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

Gov. MDMT of the Nozzle : -41 °C

Gov. MDMT of the Reinforcement Pad : -41 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -41 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C

Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $2600.00 / 5100.00 = 0.510$

Weld Size Calculations, Description: N2- shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm

Intermediate Calc. for pad/shell Welds TminPad 6.3550 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$3.1775 = 0.5 * \text{TminPad}$	$5.6560 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (16.9853 - 2.0106 + 2 \cdot 14.2748 \cdot 0.8383 \cdot (1.0 \cdot 6.355 - 5.6633)) \cdot 141)$
 $= 21730.21 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_r2) \cdot S_v$
 $= (2.8603 + 13.7259 + 1.324 - 0.0 \cdot 0.84) \cdot 141$
 $= 25705.85 \text{ Kgf}$

Weld Load [W2]:



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Nozzle Calcs.: N2- shell inlet Nozl: 11 9:53am Aug 12,2026

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (2.8603 + 0.0 + 0.8383 + (1.521)) \cdot 141$$

$$= 7491.60 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (2.8603 + 0.0 + 1.324 + 13.7259 + (1.521)) \cdot 141$$

$$= 27888.89 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 323.85 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 29992. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 463.85 \cdot 8.0 \cdot 0.49 \cdot 138$$

$$= 40191. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 154.7876) \cdot (17.4498 - 3.175) \cdot 0.7 \cdot 118$$

$$= 58465. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2.0) \cdot 323.85 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 52971. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 323.85 \cdot (9.53 - 3.175) \cdot 0.74 \cdot 141$$

$$= 34335. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (40191 + 58465) = 98656 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (29992 + 52971 + 34335 + 0) = 117298 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (40191 + 34335 + 0) = 74526 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 98656	Kgf, must exceed W = 21730 Kgf	or W1 = 25705 Kgf
Path 2-2 = 117298	Kgf, must exceed W = 21730 Kgf	or W2 = 7491 Kgf
Path 3-3 = 74526	Kgf, must exceed W = 21730 Kgf	or W3 = 27888 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2819 KPa.

The Drop for this Nozzle is : 47.4524 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 306.9824 mm

Input Echo, WRC297 Item 1, Description: N2- shell inle:

Diameter Basis for Cylindrical Shell

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Nozzle Calcs.: N2- shell inlet Nozl: 11 9:53am Aug 12,2026

Shell Internal Corrosion Allowance		3.1750	mm
Shell External Corrosion Allowance		0.0000	mm
Shell Diameter		600.000	mm
Shell Thickness		9.5300	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	161.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	140.75	N./mm ²
Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.1750	mm
Nozzle Diameter		323.850	mm
Nozzle Thickness		17.4498	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	463.850	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2600.00	KPa.
Radial Load	P	-826.02	Kgf
Circumferential Shear	Vc	1223.74	Kgf
Longitudinal Shear	Vl	1223.74	Kgf
Circumferential Moment	Mc	1.10	Kg-m.
Longitudinal Moment	Ml	1.47	Kg-m.
Torsional Moment	Mt	1.83	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Dv (.523) is > 0.5
check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{(311.353(9.53 - 3.175))}$$

$$= 470.640 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	622.705	mm
Nozzle Outside Diameter	(d) =	323.850	mm
Vessel Thickness used	(T) =	16.355	mm
Nozzle Thickness used	(t) =	14.275	mm
T / t	=	1.146	
d / t	=	22.687	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.209$$

Nr/P	=	0.052
Mr/P	=	0.090
M0/P	=	0.032
N0/P	=	0.119



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Nozzle Calcs.: N2- shell inlet

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MrD/Mc = 0.229
 NrDL/Mc = 0.096
 MØd/Mc = 0.090
 NØDL/Mc = 0.117
 MrD/Ml = 0.099
 NrDL/Ml = 0.045
 MØD/Ml = 0.039
 NØDL/Ml = 0.152

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	46	49	46	49
Outplane Stress Summary	56	47	56	47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	23	23	23	23
Inplane Stress Summary	41	8	41	8

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	56	47	56	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0



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Nozzle Calcs.: N2- shell inlet Nozl: 11 9:53am Aug 12,2026

Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	23	23	23	23

Outplane Stress Summary	32	21	33	21
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	16	-16	16	-16
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	46	49	46	49

Inplane Stress Summary	64	34	64	34
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
----------------------	----	----	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	64	35	65	34
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	30	33	30	33

Hoop Stress Summary	34	37	34	37
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0



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Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	31	0	31	0

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	35	37	35	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	30	33	30	33
Hoop Stress Summary	34	37	34	37

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	15	-15	15	-15
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	15	15	15	15
Axial Stress Summary	31	0	31	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside



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Nozzle Calcs.: N2- shell inlet Nozl: 11 9:53am Aug 12,2026

Two * Max Shear Stress 35 37 35 37

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 323.85 + 2 \cdot 1.65 \cdot \sqrt{306.352(9.53 - 3.175)}$$

$$= 469.457 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 612.705 mm
 Nozzle Outside Diameter (d) = 463.850 mm
 Vessel Thickness used (T) = 6.355 mm
 Nozzle Thickness used (t) = 84.275 mm
 T / t = 0.075
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 7.434$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.071
 M0/P = 0.019
 N0/P = 0.026
 MrD/Mc = 0.239
 NrDL/Mc = 0.091
 M0d/Mc = 0.067
 N0DL/Mc = 0.048
 MrD/Ml = 0.037
 NrDL/Ml = 0.035
 M0D/Ml = 0.012
 N0DL/Ml = 0.022

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	122	125	122	125
Outplane Stress Summary	151	107	151	107

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	84	-84	84	-84
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	155	-14	155	-14



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Nozzle Calcs.: N2- shell inlet

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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			2	2	-2	-2
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE			Au	Al	Bu	B1
(Stress Intensities)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress			156	122	156	122

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			23	-23	23	-23
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			0	0	0	0
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			61	61	61	61
Outplane Stress Summary			89	43	90	43

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stresses parallel to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)			9	9	9	9
Inplane Bending (P)			84	-84	84	-84
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			0	0	0	0
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			122	125	122	125
Inplane Stress Summary			215	50	217	48

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Shear stress normal to circumferential plane)			Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-2	-2	2	2
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-2	-2	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE			Cu	Cl	Du	D1
(Stress Intensities)			Left Outside	Left Inside	Right Outside	Right Inside



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Nozzle Calcs.: N2- shell inlet Nozl: 11 9:53am Aug 12,2026

Two * Max Shear Stress 215 51 217 49

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	46	49	46	49	46	49	46	49
Circ. Pl (SUS)	3	3	3	3	1	1	1	1
Circ. Q (SUS)	5	-5	5	-5	16	-16	16	-16
Long. Pm (SUS)	23	23	23	23	23	23	23	23
Long. Pl (SUS)	1	1	1	1	3	3	3	3
Long. Q (SUS)	16	-16	16	-16	5	-5	5	-5
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	46.0	49.0	46.0	49.0	46.0	49.0	46.0	49.0
Pm+Pl (SUS)	49.0	52.0	49.0	52.0	47.0	50.0	47.0	50.0
Pm+Pl+Q (Total)	54.1	47.0	54.1	47.0	63.0	34.1	63.0	34.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	140.75	Passed
Pm+Pl (SUS)	52.04	211.13	Passed
Pm+Pl+Q (TOTAL)	63.03	422.26	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	122	125	122	125	122	125	122	125
Circ. Pl (SUS)	5	5	5	5	9	9	9	9
Circ. Q (SUS)	23	-23	23	-23	84	-84	84	-84
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	9	9	9	9	5	5	5	5
Long. Q (SUS)	84	-84	84	-84	23	-23	23	-23
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	122.0	125.0	122.0	125.0	122.0	125.0	122.0	125.0
Pm+Pl (SUS)	127.1	130.1	127.1	130.1	131.1	134.1	131.1	134.1
Pm+Pl+Q (Total)	154.8	121.1	154.8	121.1	215.0	50.5	215.0	50.5



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	125.00	140.75	Passed
Pm+P1 (SUS)	134.06	211.13	Passed
Pm+P1+Q (TOTAL)	215.03	422.26	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	33	30	33	30	33	30	33
Circ. P1 (SUS)	3	3	3	3	3	3	3	3
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. P1 (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	15	-15	15	-15	15	-15	15	-15
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	30.0	33.0	30.0	33.0	30.0	33.0	30.0	33.0
Pm+P1 (SUS)	33.1	36.0	33.1	36.0	33.1	36.0	33.1	36.0
Pm+P1+Q (Total)	33.3	36.1	33.3	36.1	33.3	36.1	33.3	36.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	118.00	Passed
Pm+P1 (SUS)	36.05	177.00	Passed
Pm+P1+Q (TOTAL)	36.06	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: N2- Channel out

Nozl: 12

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Input, Nozzle Desc: N2- Channel outlet

From: 60

Pressure for Reinforcement Calculations	P	2000.000	KPa.
Temperature for Internal Pressure	Temp	66	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	600.00	mm
Shell Finished (Minimum) Thickness	t	9.5300	mm
Shell Internal Corrosion Allowance	c	3.1750	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		6053.80	mm
User Entered Minimum Design Metal Temperature		11.40	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.1750	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	9.5300	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	308.2750	mm
Thickness of Pad	te	10.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	10.0000	mm
Reinforcing Pad Width		70.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



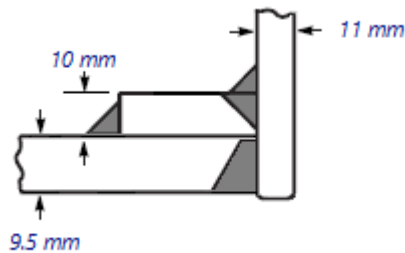
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Nozzle Calcs.: N2- Channel out

Noz1: 12 9:53am Aug 12,2026



Reinforcement CALCULATION, Description: N2- Channel outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	6.625 in.
Actual Thickness Used in Calculation	0.432 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2000.0 \cdot 303.175 / (118 \cdot 1.0 - 0.6 \cdot 2000.0)$
 $= 5.1918 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 2000.0 \cdot 84.1375 / (118 \cdot 1.0 + 0.4 \cdot 2000.0)$
 $= 1.4166 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	305.3588 mm
Parallel to Vessel Wall, opening length	d	152.6794 mm
Normal to Vessel Wall (Thickness Limit), pad side TLwp		15.8875 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

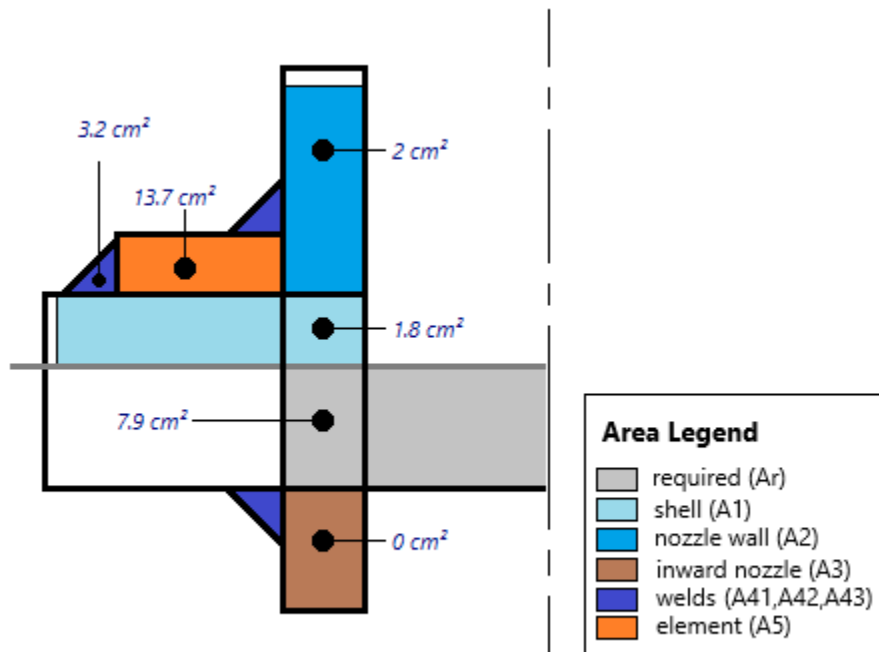


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	7.927	NA	NA
Area in Shell	A1	1.776	NA	NA
Area in Nozzle Wall	A2	2.028	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	3.186	NA	NA
Area in Element	A5	13.708	NA	NA
TOTAL AREA AVAILABLE	Atot	20.698	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (152.6794 \cdot 5.1918 \cdot 1.0 + 2 \cdot 7.7978 \cdot 5.1918 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 7.927 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 152.679(1.0 \cdot 6.355 - 1.0 \cdot 5.192) - 2 \cdot 7.798 \\
 &\quad (1.0 \cdot 6.355 - 1.0 \cdot 5.192) \cdot (1 - 1.0) \\
 &= 1.776 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 15.89) \cdot (7.8 - 1.42) \cdot 1.0
 \end{aligned}$$



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

= 2.028 cm²

Area Available in Welds [A41 + A42 + A43]:

= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4

= (3.1859) • 1.0 + (0.0) • 1.0 + 0.0² • 1.0

= 3.186 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4

= (305.3588 - 168.275) • 10.0 • 1.0

= 13.708 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 4.5916 mm

Wall Thickness per UG16(b), tr16b = 5.6750 mm

Wall Thickness, shell/head, internal pressure trb1 = 8.3668 mm

Wall Thickness tb1 = max(trb1, tr16b) = 8.3668 mm

Wall Thickness tb2 = max(trb2, tr16b) = 5.6750 mm

Wall Thickness per table UG-45 tb3 = 9.3950 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]

= min[9.395, max(8.3668, 5.675)]

= 8.3668 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(4.5916, 8.3668)

= 8.3668 mm

Available Nozzle Neck Thickness = 0.875 * 10.973 = 9.601 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 8.3, Allowable : 118.0 N./mm² Passed

Expansion : 0.0, Allowable : 286.7 N./mm² Passed

Occasional : 9.3, Allowable : 156.9 N./mm² Passed

Shear : 4.6, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.601, tr = 1.417, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.22, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.601, tr = 1.417, c = 3.175 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.22, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 1.417, c = 3.175 mm, E* = 1.0



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

Thickness Ratio = $tr * E^* / (tg - c) = 0.22$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 9.53$, $tr = 5.192$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (tg - c) = 0.817$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 9.53$, $tr = 5.192$, $c = 3.175$ mm, $E^* = 1.0$
Thickness Ratio = $tr * E^* / (tg - c) = 0.817$, Temp. Reduction = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

Gov. MDMT of the Nozzle : -39 °C
Gov. MDMT of the Reinforcement Pad : -39 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $2000.00 / 5100.00 = 0.392$

Weld Size Calculations, Description: N2- Channel out

Intermediate Calc. for nozzle/shell Welds T_{min} 7.7978 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 6.3550 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$5.4585 = 0.7 * t_{min}$	$21.2100 = 0.7 * W_o$ mm
Pad Weld	$3.1775 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (7.9267 - 1.776 + 2 \cdot 7.7978 \cdot 1.0 \cdot (1.0 \cdot 6.355 - 5.1918)) \cdot 118)$
 $= 7618.93$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot Sv$
 $= (2.0276 + 13.7084 + 3.1859 - 0.0 \cdot 1.0) \cdot 118$
 $= 22767.22$ Kg

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot Sv$
 $= (2.0276 + 0.0 + 7.0084 + (0.9911)) \cdot 118$
 $= 12064.83$ Kg

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

$$= (2.0276 + 0.0 + 3.1859 + 13.7084 + (0.9911)) \cdot 118$$

$$= 23959.74 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 30.0 \cdot 0.49 \cdot 118$$

$$= 46752. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 308.275 \cdot 8.0 \cdot 0.49 \cdot 118$$

$$= 22840. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 80.2386) \cdot (10.9728 - 3.175) \cdot 0.7 \cdot 118$$

$$= 16556. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 168.275 \cdot 10.0 \cdot 0.74 \cdot 138$$

$$= 27524. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (9.53 - 3.175) \cdot 0.74 \cdot 118$$

$$= 14957. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (22840 + 16556) = 39395 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (46752 + 27524 + 14957 + 0) = 89233 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (22840 + 14957 + 0) = 37796 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 39395	Kgf, must exceed W = 7618 Kgf	or W1 = 22767 Kgf
Path 2-2 = 89232	Kgf, must exceed W = 7618 Kgf	or W2 = 12064 Kgf
Path 3-3 = 37796	Kgf, must exceed W = 7618 Kgf	or W3 = 23959 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2437 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.0402 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 271.5701 mm

Input Echo, WRC297 Item 1, Description: N2- Channel ou:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.1750 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	600.000 mm
Shell Thickness	9.5300 mm



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

Shell Stress Concentration Factor		1.000	
Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.1750	mm
Nozzle Diameter		168.275	mm
Nozzle Thickness		10.9728	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	308.275	mm
Thickness of Reinforcing Pad	10.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2000.00	KPa.
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	0.28	Kg-m.
Longitudinal Moment	Ml	0.37	Kg-m.
Torsional Moment	Mt	0.46	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(311.353(9.53 - 3.175))}$$

$$= 315.065 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	622.705	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	16.355	mm
Nozzle Thickness used	(t) =	7.798	mm
T / t	=	2.097	
d / t	=	21.580	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.667$$

Nr/P	=	0.077
Mr/P	=	0.087
M0/P	=	0.053
N0/P	=	0.154
MrD/Mc	=	0.196
NrDL/Mc	=	0.131
M0d/Mc	=	0.142



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Nozzle Calcs.: N2- Channel out

Nozl: 12 9:53am Aug 12,2026

NØDL/Mc = 0.151
MrD/Ml = 0.134
NrDL/Ml = 0.085
MØD/Ml = 0.093
NØDL/Ml = 0.266

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	36	38	36	38
Outplane Stress Summary	43	35	43	35

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	18	18	18	18
Inplane Stress Summary	26	11	27	11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	43	35	43	35

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

Normal Pressure Stress	18	18	18	18
------------------------	----	----	----	----

Outplane Stress Summary	25	15	25	15
-------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	7	-7	7	-7
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	36	38	36	38

Inplane Stress Summary	44	31	45	31
------------------------	----	----	----	----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
----------------------	----	----	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	45	31	45	31
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24

Hoop Stress Summary	24	27	24	27
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

Axial Stress Summary	31	-6	31	-7
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	2	2	-2	-2
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	32	34	32	34
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	22	24	22	24

Hoop Stress Summary	24	27	24	27
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	19	-19	19	-19
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	11	11	11	11

Axial Stress Summary	31	-6	31	-7
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	0	0	0	0

Shear Stress Summary	-2	-2	2	2
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	32	34	32	34
------------------------	----	----	----	----



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{306.352(9.53 - 3.175)}$$

$$= 313.882 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 612.705 mm
 Nozzle Outside Diameter (d) = 308.275 mm
 Vessel Thickness used (T) = 6.355 mm
 Nozzle Thickness used (t) = 77.798 mm
 T / t = 0.082
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.940$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.071
 M0/P = 0.020
 N0/P = 0.027
 MrD/Mc = 0.241
 NrDL/Mc = 0.099
 M0d/Mc = 0.072
 N0DL/Mc = 0.053
 MrD/M1 = 0.050
 NrDL/M1 = 0.048
 M0D/M1 = 0.015
 N0DL/M1 = 0.027

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	94	96	94	96
Outplane Stress Summary	108	87	108	87

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	41	-41	41	-41
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	47	47	47	47
Inplane Stress Summary	93	9	93	9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			0	0	0	0
Shear Stress Summary			1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress			108	87	109	87

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)			2	2	2	2
Outplane Bending (P)			11	-11	11	-11
Outplane Membrane (Mc)			0	0	0	0
Outplane Bending (Mc)			0	0	0	0
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			47	47	47	47
Outplane Stress Summary			61	38	61	38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)			4	4	4	4
Inplane Bending (P)			41	-41	41	-41
Inplane Membrane (Mc)			0	0	0	0
Inplane Bending (Mc)			0	0	0	0
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			94	96	94	96
Inplane Stress Summary			140	59	141	58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			0	0	0	0
Shear Stress Summary			-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress			140	59	141	58



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Nozzle Calcs.: N2- Channel out

Nozl: 12 9:53am Aug 12,2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	36	38	36	38	36	38	36	38
Circ. Pl (SUS)	2	2	2	2	1	1	1	1
Circ. Q (SUS)	4	-4	4	-4	7	-7	7	-7
Long. Pm (SUS)	18	18	18	18	18	18	18	18
Long. Pl (SUS)	1	1	1	1	2	2	2	2
Long. Q (SUS)	7	-7	7	-7	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	36.0	38.0	36.0	38.0	36.0	38.0	36.0	38.0
Pm+Pl (SUS)	38.1	40.0	38.1	40.0	37.1	39.1	37.1	39.1
Pm+Pl+Q (Total)	42.1	36.0	42.1	36.0	44.0	32.1	44.0	32.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	38.00	118.00	Passed
Pm+Pl (SUS)	40.05	177.00	Passed
Pm+Pl+Q (TOTAL)	44.05	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	94	96	94	96	94	96	94	96
Circ. Pl (SUS)	2	2	2	2	4	4	4	4
Circ. Q (SUS)	11	-11	11	-11	41	-41	41	-41
Long. Pm (SUS)	47	47	47	47	47	47	47	47
Long. Pl (SUS)	4	4	4	4	2	2	2	2
Long. Q (SUS)	41	-41	41	-41	11	-11	11	-11
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	94.0	96.0	94.0	96.0	94.0	96.0	94.0	96.0
Pm+Pl (SUS)	96.0	98.0	96.0	98.0	98.0	100.0	98.0	100.0
Pm+Pl+Q (Total)	107.1	87.0	107.1	87.0	139.0	59.0	139.0	59.0

Vessel Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N2- Channel out Nozl: 12 9:53am Aug 12,2026

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	96.00	118.00	Passed
Pm+P1 (SUS)	100.02	177.00	Passed
Pm+P1+Q (TOTAL)	139.01	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	22	24	22	24	22	24	22	24
Circ. P1 (SUS)	2	2	2	2	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. P1 (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	19	-19	19	-19	19	-19	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	22.0	24.0	22.0	24.0	22.0	24.0	22.0	24.0
Pm+P1 (SUS)	24.3	26.3	24.3	26.3	24.3	26.3	24.3	26.3
Pm+P1+Q (Total)	31.5	33.2	31.5	33.2	31.5	33.2	31.5	33.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	118.00	Passed
Pm+P1 (SUS)	26.28	177.00	Passed
Pm+P1+Q (TOTAL)	33.24	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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ASME TS Calc: DESIGN CASE

Case: 1 9:53a Aug 12, 2026

Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	2600.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2600.00 KPa.
Shell Minimum Operating Pressure	Psox,min	2220.00 KPa.
Shell Thickness	ts	9.5300 mm
Shell Internal Corrosion Allowance	cas	3.1750 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	600.000 mm
Shell Circumferential Joint Efficiency	Esw	1.000
Shell Temperature for Internal Pressure	Ts	165.56 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	2000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	2000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	441.30 KPa.
Channel Thickness	tc	9.5300 mm
Channel Internal Corrosion Allowance	cac	3.1750 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	600.000 mm
Channel Design Temperature	TEMPC	65.56 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	583
Tube Wall Thickness	et	2.1080 mm
Tube Outside Diameter	D	15.8750 mm
Total Straight Tube Length	Lt	4877.00 mm
Straight Tube Length (bet. inner tubsht faces) L		4757.00 mm
Design Temperature of the Tubes		165.56 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	156.75 N./mm ²
Tube Pitch (Center to Center Spacing)	P	19.8440 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.0000 mm
Groove Weld Leg	ag	2.0000 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	285.351 mm



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ASME TS Calc: DESIGN CASE

Case: 1 9:53a Aug 12, 2026

Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	1839.8 cm ²
Perimeter of Tube Layout	Cp	1715.8 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature	T	165.56 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	60.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.1750 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.1750 mm
Tubesheet Outside Diameter	A	740.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	663.00 mm
Step 1 Depth on the Tubeside	ht1	8.00 mm
Step 2 Diameter on the Tubeside	dt2	597.00 mm
Step 2 Depth on the Tubeside	ht2	8.00 mm
Is the Tubesheet Flat on Shellside?		No
Calculated Tubesheet Diameter per Div 2, 4.18.4		663.00 mm

Note: Tubesheet diameter is now: 663.000 mm Div 2, 4.18.4.

Area of the Untubed Lanes	AL	0.0 cm ²
---------------------------	----	---------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Mean Shell Metal Temp. along Shell len.	Tsm	70.57 °C
Mean Tube Metal Temp. along Tube length	Ttm	50.30 °C
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	740.000 mm
Flange Inside Diameter	B	600.000 mm
Flange Face Outside Diameter	Fod	666.000 mm
Flange Face Inside Diameter	Fid	600.000 mm
Gasket Outside Diameter	Go	655.000 mm
Gasket Inside Diameter	Gi	629.000 mm
Small end Hub thk.	g0	9.5300 mm
Large end Hub thk.	g1	13.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects



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ASME TS Calc: DESIGN CASE

Case: 1 9:53a Aug 12,2026

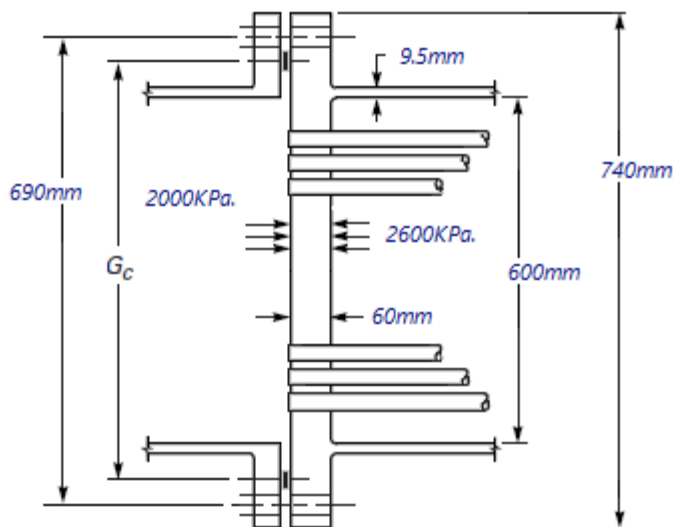
Bolting Information:

Diameter of Bolt Circle	C	690.000 mm
Nominal Bolt Diameter	dB	19.0500 mm
Thread Series	TEMA Thread Series	
Number of Bolts	n	28
Bolt Material	SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Tubesheet Integral with	Shell	
Tubesheet Extended as Flange	Yes	
Thickness of Extended Portion of Tubesheet	Tf	44.0000 mm
Is Bolt Load Transferred to the Tubesheet	Yes	

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	70.6 °C	0.0000119115 /°C
Elastic Mod. at Design Temperature	165.6 °C	0.19429E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	70.6 °C	0.19987E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%



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ASME TS Calc: DESIGN CASE

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Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	65.6 °C	0.20016E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Carbon & Low Alloy Steels, Group 1
Tubes - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	50.3 °C	0.0000117802 /°C
Elastic Mod. at Design Temperature	165.6 °C	0.19429E+09 KPa.
Elastic Mod. at Metal Temp. along Len	50.3 °C	0.20103E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	165.6 °C	0.19429E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	165.6 °C	0.19429E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (2000.0 \cdot (15.875/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 2000.0)$$

$$= 0.1703 + 0.0000 = 0.1703 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 165.56 °C
Elastic Modulus for Material 197876288.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1080	15.88	4757.00	7.53	50.0000	0.0193957	95.01

EMAWP = (2.167/(D/T) - 0.0833) * B = 19424.4336 KPa.

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4417	15.88	4757.00	35.94	50.0000	0.0008516	70.09

EMAWP = (4*B)/(3*(D/T)) = (4 * 70.0947)/(3 * 35.9409) = 2600.2195 KPa.

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4417 mm
Allowable Internal Pressure at Corroded thickness	27453.64 KPa.
Required Internal Design Pressure	2000.00 KPa.
Allowable External Pressure at Corroded thickness	19424.43 KPa.
Required External Design Pressure	2600.00 KPa.
Required Thickness due to Shell Side pressure	0.4417 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	N = (Goc-Gic) / 2	13.000 mm
Basic Gasket Width,	b0 = N / 2	6.500 mm



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ASME TS Calc: DESIGN CASE

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Effective Gasket Width, $b = \sqrt{b_0} * 2.5$ 6.422 mm
Gasket Reaction Diameter, $G = G_0 - 2.0 * b$ 642.156 mm

Results for ASME Fixed Tubesheet Calculations for Configuration b:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio $[\rho]$:

= l_{tx} / h (modified for corrosion if present)

= 0/60.0

= 0.0000 (must be $0 \leq \rho \leq 1$)

Effective tube hole diameter $[d^*]$:

= $\max(dt - 2 * tt(E_t T / E) * (S_t T / S) * (\rho); dt - 2 * tt)$

= $\max(15.875 - 2 * 2.108 (0.19429E+09 / 0.19429E+09) * (92/137) * (0); 15.875 - 2 * 2.108)$

= 15.8750 mm

Equivalent outer tube limit circle diameter $[Do]$:

= $2 * r_o + dt$

= $2 * 285.351 + 15.875$

= 586.577 mm

Determine the basic ligament efficiency for shear $[\mu]$:

= $(p - dt) / p$

= $(19.844 - 15.875) / 19.844$

= 0.200

Equivalent outer tube limit radius $[ao]$:

= $Do / 2$

= 586.577/2

= 293.2885 mm

Effective tube pitch $[p^*]$:

= $p / \sqrt{ 1 - 4 * \min(AL * CNVfactor; 4 * Do * p) / (\pi * Do^2) }$

= $19.844 / \sqrt{ 1 - 4 * \min(0 * 100.0 ; 4 * 586.577 * 19.844) / (\pi * 586.577^2) }$

= 19.8440 mm

Effective ligament efficiency for bending $[\mu^*]$:

= $(p^* - d^*) / p^*$

= $(19.844 - 15.875) / 19.844$

= 0.2000

Ratio $[ps]$:

= as / ao

= 300.0/293.2885

= 1.0229

Ratio $[pc]$:

= ac / ao

= 321.0778/293.2885

= 1.0948

Compute parameter $[xt]$:

= $1 - Nt((dt - 2 * tt) / (2 * ao))^2$

= $1 - 583 ((15.875 - 2 * 2.108) / (2 * 293.2885))^2$

= 0.7697

Determine parameter $[xs]$:

= $1 - Nt(dt / (2 * ao))^2$

= $1 - 583 (15.875 / (2 * 293.2885))^2$

= 0.5730



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ASME TS Calc: DESIGN CASE

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Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot ts(D_s + ts)E_s/L$$

$$= \pi \cdot 9.53 (600.0 + 9.53) \cdot 1.9429E+09/4757.0$$

$$= 76006.7891 \text{ Kg/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot tt(D_t - tt)E_t/L$$

$$= \pi \cdot 2.108 (15.875 - 2.108) \cdot 1.9429E+09/4757.0$$

$$= 379.7295 \text{ Kg/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (N_t \cdot K_t)$$

$$= 76006.789 / (583 \cdot 379.729)$$

$$= 0.34333$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s/K_j)$$

$$= 1 / (1 + 76006.789/0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [βs]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + ts) ts)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((600.0 + 9.53) 9.53)$$

$$= 0.0239 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot ts^3 / (6(1 - \nu_s^2))$$

$$= 0.024 \cdot 0.19429E+09 \cdot 9.53^3 / (6(1 - 0.3^2))$$

$$= 734596544.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [λs]:

$$= 6(D_s + ts) \cdot ks/h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2/2)$$

$$= 6(600.0 + 9.53) \cdot 0.73460E+09/60.0^3 \cdot (1 + 60.0 \cdot 0.024 + (60.0 \cdot 0.024)^2/2)$$

$$= 42973220.0000 \text{ KPa}$$

Shell coefficient [δs]:

$$= D_s^2 / (4 \cdot E_s \cdot ts) \cdot (1 - D_s \cdot \nu_s / (2(D_s + ts)))$$

$$= 600.0^2 / (4 \cdot 0.19429E+09 \cdot 9.53) \cdot (1 - 600.0 \cdot 0.3 / (2(600.0 + 9.53)))$$

$$= 0.41428E-01 \text{ mm/N./mm}^2$$

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

βC, kc, λC, δC are all = 0

ASME Section VIII-2 Part 4.18.8.4 Step 3:

h/p = 3.023584 ; μ* = 0.200010 ; E* = 30185138 KPa.

E*/E = 0.155361 ; ν* = 0.458557

Note: As h/p (3.024) is > 2, data values for h/p = 2 were used.

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^{*2})N_t \cdot E_t \cdot tt(d_t - tt)a_o^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.459^2) 583 \cdot 0.19429E+09 \cdot 2.108$$

$$(15.875 - 2.108) 293.2885^2) / (30185138 \cdot 4757.0 \cdot 60.0^3))^{1/4}$$

$$= 3.6256$$

Values from Table 4.18.3



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Zd = 0.031494 ; Zv = 0.073484 ; Zm = 0.394465
Za = 0.455587E+01 ; Zw = 0.073484

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

= A / Do
= 663.0/586.577
= 1.1303

Compute coefficient [F]:

= (1 - v*) / E* * (λs + λc + E • ln(K))
= (1 - 0.46) / 30185138 • (42973220 + 0 +
0.19429E+09 • ln(1.13))
= 1.1976

Compute parameter [φ]:

= (1 + v*) F
= (1 + 0.4586) 1.1976
= 1.7468

Compute parameter [Q1]:

= (ps - 1 - φ • Zv) / (1 + φ • Zm)
= (1.0229 - 1 - 1.7468 • 0.0735) / (1 + 1.7468 • 0.3945)
= -0.062449

Compute Parameter [Qz1]:

= (Zd + Q1 • Zw) / 2 • Xa⁴
= (0.03149 + -0.06245 • 0.07348) / 2 • 3.62562⁴
= 2.3245

Compute parameter [Qz2]:

= (Zv + Q1 • Zm) / 2 • Xa⁴
= (0.07348 + -0.06245 • 0.39447) / 2 • 3.62562⁴
= 4.2205

Compute parameter [U]:

= (Zw + (ps - 1) Zm) Xa⁴ / (1 + φ • Zm)
= (0.0735 + (1.0229 - 1) 0.3945) 3.62562⁴ / (1 + 1.7468 • 0.3945)
= 8.4410

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [yb]:

= (Gc - C) / Do (config b)
= (642.1556 - 690.0) / 586.577 = -0.08157

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ws]:

= ps • ks • βs • δS (1 + h • βs)
= 1.0229 • 0.73460E+09 • 0.0239 • 0.041428 (1 + 60.0 • 0.0239)
= 1805.0922 mm²

Calculate parameter [ω*s]:

= ao² (ps² - 1) • (ps - 1) / 4 - ωs
= 293.289² (1.023² - 1) • (1.023 - 1) / 4 - 1805.092
= -1782.3125 mm²

Parameter [ωc]:

= pc • kc [δC (CmTheta • βC + CqΔ • h • βC²) -
ζc (CmTheta + h • CqTheta • h • βC / 2))
= 1.095 • 0 [0 (0 + 0 • 0 • 60.0) -
0 (0/60.0 + 0 • 0/2))



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$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - \omega c$$

$$= 293.28851^2 [(1.09475^2 + 1) \cdot (1.09475 - 1)/4 - (1.02288 - 1)/2] - 0$$

$$= 3495.3523 \text{ mm}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [$P's$]:

$$= P_s \{ x_s + 2(1 - x_s) v_t + [2/Kst \cdot (D_s/D_o)^2] v_s - [(p_s^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(D_j^2 - (D_s)^2)/D_o^2] \}$$

$$= 2600.0 \{ 0.573 + 2(1 - 0.573) 0.3 + [2/0.343 \cdot (600.0/586.577)^2] 0.3 - [(1.023^2 - 1)/(1.0 \cdot 0.343)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.343)] [(0^2 - (600.0)^2)/586.577^2] \}$$

$$= 6559.4360 \text{ KPa.}$$

Pressure [$P't$]:

$$= [x_t + 2(1 - x_t) v_t + 1/(J \cdot Kst)] P_t$$

$$= [0.77 + 2(1 - 0.77) 0.3 + 1/(1.0 \cdot 0.343)] 2000.0$$

$$= 7641.0693 \text{ KPa.}$$

Pressure [P_y]:

$$= N_t \cdot K_t \cdot \gamma / (\pi \cdot a_o^2)$$

$$= 583 \cdot 379.729 \cdot 0 / (3.142 \cdot 293.289^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [P_w]:

$$= -\gamma_b \cdot U \cdot W \cdot / (2 \cdot \pi \cdot a_o^2)$$

$$= -0.082 \cdot 8.441 \cdot 81908.16 / (2 \cdot \pi \cdot 293.289^2)$$

$$= 1023.1957 \text{ KPa.}$$

Calculate the pressure [$Prim$]:

$$= -(U/a_o^2) \cdot (\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t)$$

$$= -(8.441/293.289^2) \cdot (-1782.312 \cdot 2600.0 - 3495.352 \cdot 2000.0)$$

$$= 1140.7321 \text{ KPa.}$$

Calculate the pressure [P_w]:

$$= (U/a_o^2) \cdot (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)$$

$$= (8.441/293.289^2) \cdot (1805.0922 \cdot 0 - 0 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (p_s - 1)Qz2)) \cdot (P's - P't + P_y + P_w + P_w + Prim)$$

$$= 0.1000E+01 \cdot 0.343 / (1 + 1.0 \cdot 0.343 (2.324 + (1.023 - 1)4.22)) \cdot (6559.436 - 7641.069 + 0 + 0 + 1023.196 + 1140.732)$$

$$= 202.9152 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [$Q2$]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((-1782.312 \cdot 2600.0 - 3495.352 \cdot 2000.0) - (1805.092 \cdot 0 - 0 \cdot 0)) \cdot 0 + 81908.2 \cdot -0.082 / (2 \cdot \pi)] / (1 + 1.74684 \cdot 0.39447)$$

$$= -1331.3467 \text{ Kgf}$$



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Calculate factor [Q3]:

$$\begin{aligned}
 &= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2) \\
 &= -0.062 + 2 \cdot -1331.347 / (202.915 \cdot 293.289^2) \\
 &= -1.55841
 \end{aligned}$$

Fm Value from Table 4.18.3 = 0.779204

Tubesheet Bending Stress - original thickness [σ]:

$$\begin{aligned}
 &= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.779204) \cdot (2 \cdot 293.2885 / (60.0 - 0))^2 \cdot 202.92 \\
 &= 113.3387 \text{ N./mm}^2
 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138.0 = 207.0 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$\begin{aligned}
 &= (1.5 \cdot Fm / \mu \cdot) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe \\
 &= (1.5 \cdot 0.3788 / 0.2) \cdot (2 \cdot 293.2885 / (41.867 - 0))^2 \cdot 371.14 \\
 &= 206.9821 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$\begin{aligned}
 &= h + Cats + Catc \\
 &= 41.8671 + 0 + 0 \\
 &= 41.8671 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$\begin{aligned}
 &= (2 \cdot \mu \cdot h / ao) \cdot \min(0.8S; 0.533Sy) \\
 &= (2 \cdot 0.2 \cdot 60.0 / 293.289) \cdot 110.398 \\
 &= 9.0344 \text{ N./mm}^2
 \end{aligned}$$

Shear Stress is not required to be computed; [Pe] <= Tau limit

Tubesheet Shear Stress is probably low, use the following req. thk:

$$\begin{aligned}
 \text{Tubesheet thickness (Incl. Corr.)} &= 3.8100 \text{ mm} \\
 \text{Tubesheet Shear Stress} &= 44.5912 \text{ N./mm}^2
 \end{aligned}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$\begin{aligned}
 &= \max(HreqB; HreqS) \\
 &= \max(41.8671; 3.81) \\
 &= 41.8671 \text{ mm}
 \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$Ftmin = -7.173, Ftmax = 5.1065$$

First Extreme Tube Axial Stress from among all the tubes [σ_1]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmin) / (Xt - Xs) \\
 &= ((2600.0 \cdot 0.573 - 2000.0 \cdot 0.7697) - (202.915) \cdot -7.173) / \\
 &\quad (0.7697 - 0.573) \\
 &= 7.1482 \text{ N./mm}^2
 \end{aligned}$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_2]:

$$\begin{aligned}
 &= ((Ps \cdot xs - Pt \cdot xt) - Pe \cdot Ftmax) / (Xt - Xs) \\
 &= ((2600.0 \cdot 0.573 - 2000.0 \cdot 0.7697) - (202.915) \cdot 5.107) / \\
 &\quad (0.7697 - 0.573) \\
 &= -5.5205 \text{ N./mm}^2
 \end{aligned}$$

Maximum Tube Axial Stress [σ ,max]:

$$= \max(|\sigma_1|; |\sigma_2|) = 7.148 \text{ N./mm}^2$$

Allowable Tube Stress [σ ,A]:



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$$= Sot = 92.3983 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)Xa^4); 1.25)$$

$$= \max((3.25 - 0.25(0.031 + -1.558 \cdot 0.073)3.626^4); 1.25)$$

$$= 2.0000 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (15.875^2 + (15.875 - 2 \cdot 2.108)^2) }) / 4$$

$$= 4.9241 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19429E+09 / 156)}$$

$$= 156.4208$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 4.924$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (Sy, t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (156 / 2.0) \cdot (1 - 0 / (2 \cdot 156.421))$$

$$= 78.377 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t, \max} \cdot \text{Tube Area}$$

$$= 7.15 \cdot 0.9117$$

$$= 66.46 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (15.875 - 2.108) 2.108$$

$$= 0.912 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.108 (15.875 - 2.108) 92.398$$

$$= 859.004 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.0 (15.875 + 0.67 \cdot 2.0) 92.4$$

$$= 859.0042 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 92.398 / 92.398$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.0 (15.875 + 0.67 \cdot 2.0) 92.4$$

$$= 560.5122 \text{ Kgf}$$



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Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 15.875)^2 + 1.07 \cdot 2.108 \cdot (15.875 - 2.108) \cdot 1.0 \cdot 1.0} - 0.75 \cdot 15.875 \right)$$

$$= 2.479 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.2648 mm

Minimum Required Groove Weld Leg [agr]: 1.2648 mm

Tube-Tubesheet Jt allowable, 859.0 is >= tube strength 859.0 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 118.0 \cdot 1.0 = 118.0 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 293.289^2 / ((600.0 + 9.53) 9.53) \cdot [202.92 + (1.023^2 - 1) \cdot (2600.0 - 2000.0)] + 300.0^2 \cdot 2000.0 / ((600.0 + 9.53) 9.53)$$

$$= 34.4055 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 9.53 mm, the minimum Shell length adjacent to the tubesheet should be 136.111 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a_o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a_s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 293.289^2 / ((600.0 + 9.53) 9.53) [202.92 + (1.023^2 - 1) \cdot (2600.0 - 2000.0)] + 300.0^2 \cdot 2000.0 / ((600.0 + 9.53) 9.53)$$

$$= 34.4055 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a_s^2 \cdot P \cdot s / (E \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a_o / h)^3 [1 + h \cdot \beta_s / 2] [P_e (Z_v + Z_m \cdot Q_1) + 2 / a_o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.73460 \text{E}+09 / 9.53^2 \{ 0.024 [0.041428 \cdot 2600.0 + 300.0^2 \cdot 0 / (1.18516 \text{E}+1)] + 6(1 - 0.46^2) / (30185138) (293.29 / 60.0)^3 [1 + 60.0 \cdot 0.02 / 2] [202.9 (0.073 + 0.394 \cdot -0.062) + 2 / 293.29^2 \cdot 0.394 \cdot -1331.347] \}$$

$$= -42.9650 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_{s,m}| + |\sigma_{s,b}| \leq 1.5 \cdot S_s$$

$$\therefore |34.41| + |-42.97| \leq 177.0 \text{ N./mm}^2$$

$$\therefore 77.37, \text{ must be } \leq 177.0 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: b

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	113.3 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	1.8 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	7.1 <=	92.4 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-5.5 <=	-78.4 N./mm ²	Pass
Maximum Force on any one Tube	66.5 <=	859.0 Kgf	Pass
Axial Membrane Stress in Shell	34.4 <=	118.0 N./mm ²	Pass



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Shell Stress (jt. inter.) 77.4 <= 177.0 N./mm² Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	41.867	60.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.265	2.000	Ok
Tube-Tubesheet Groove Weld Leg :	1.265	2.000	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 14 Load Cases:

	--Reqd. Thk. + CA		----	Tubesheet	Stresses		Case	Pass/	
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail	
D1uc	34.761	21.674	102	207	10	110	Ps+Pt-Th	D1	Ok
D2uc	22.213	...	52	207	12	110	Ps+Pt-Th	D2	Ok
D3uc	41.867	...	113	207	2	110	Ps+Pt-Th	D3	Ok
O1uc	17.971	...	162	434	25	110	Ps+Pt+Th	O1	Ok
O2uc	16.301	...	80	434	2	110	Ps+Pt+Th	O2	Ok
O3uc	20.622	...	137	434	14	110	Ps+Pt+Th	O3	Ok
O4uc	4.756	...	93	434	14	110	Ps+Pt+Th	O4	Ok
D1c	49.285	21.674	146	207	11	110	Ps+Pt-Th-c	D1	Ok
D2c	23.064	...	49	207	12	110	Ps+Pt-Th-c	D2	Ok
D3c	50.361	...	144	207	1	110	Ps+Pt-Th-c	D3	Ok
O1c	36.421	...	203	434	24	110	Ps+Pt+Th-c	O1	Ok
O2c	30.598	...	112	434	2	110	Ps+Pt+Th-c	O2	Ok
O3c	35.845	...	174	434	11	110	Ps+Pt+Th-c	O3	Ok
O4c	26.725	...	113	434	10	110	Ps+Pt+Th-c	O4	Ok
Max:	50.3605	21.674	mm	0.707		0.230	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	13	117	...	...	:	...	...	...	...	:	Ok	



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D2uc	21	117	...	...	...	...	...	...	Ok
D3uc	34	117	...	...	...	...	...	...	Ok
O1uc	12	423	-12	-111	...	...	...	...	Ok
O2uc	2	423	-2	-111	...	...	...	...	Ok
O3uc	9	423	...	...	...	...	...	...	Ok
O4uc	23	423	-23	-111	...	...	...	...	Ok
D1c	20	117	...	...	...	...	...	...	Ok
D2c	31	117	...	...	...	...	...	...	Ok
D3c	51	117	...	...	...	...	...	...	Ok
O1c	8	423	-8	-105	...	...	...	...	Ok
O2c	8	423	...	...	...	...	...	...	Ok
O3c	23	423	...	...	...	...	...	...	Ok
O4c	23	423	-23	-105	...	...	...	...	Ok

Max RATIO 0.433 0.221 ...

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses		Tube Loads		Shell Stress		Channel Stress		Stress Pass	
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd Fail
D1uc	14	92	-14	-92	134	859	240	423	...	Ok
D2uc	9	92	-6	-78	83	859	206	423	...	Ok
D3uc	7	92	-6	-78	66	859	77	176	...	Ok
O1uc	34	184	-18	-87	318	1718	381	423	...	Ok
O2uc	18	184	...	...	163	1718	30	423	...	Ok
O3uc	28	184	-9	-92	263	1718	193	423	...	Ok
O4uc	23	184	-6	-89	218	1718	236	423	...	Ok
D1c	18	92	-16	-92	166	859	296	423	...	Ok
D2c	10	92	-8	-78	96	859	304	423	...	Ok
D3c	10	92	-6	-78	94	859	55	176	...	Ok
O1c	36	184	-20	-92	337	1718	416	423	...	Ok
O2c	14	184	...	...	132	1718	58	423	...	Ok
O3c	28	184	-10	-92	264	1718	160	423	...	Ok
O4c	22	184	-7	-92	204	1718	245	423	...	Ok

Max RATIO 0.196 0.218 0.196 0.984 ...

Summary of Thickness Comparisons for 14 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	50.361	60.000	Ok
Tubesheet thickness flanged extension :	21.674	44.000	Ok
Tube thickness :	0.442	2.108	Ok
Tube-tubesheet fillet weld leg :	1.265	2.000	Ok
Tube-tubesheet groove weld leg :	1.265	2.000	Ok

Min Shell length of thk, (9.530) adj. to tubesheet: 136.111 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.168 mm
 Due to Pressure Shell Compresses by : -0.066 mm
 Due to Pressure + Thermal Shell Compresses by : -1.235 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	2887.01	1.000	10290.94	1.000
Tubesheet Shear Stress	16247.52	1.000	23031.59	1.000
Tube Tensile Stress	10357.99	1.000	23231.76	1.000



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ASME TS Calc: DESIGN CASE

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Tube Compressive Stress	6029.17	0.538	26181.85	1.000
Tube-Tubesheet Joint load	10358.00	1.000	23231.76	1.000
Shell Stress (Axial, Junction)	2068.00	0.998	3623.53	1.000
Tube Pressure Stress	27453.63	1.000	19424.42	1.000
Tubesheet Extension Stress	2336.28	...	No Calc	No Calc
Minimum MAWP	2068.00		3623.53	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	4106.98	1.000	10290.94	1.000
Tubesheet Shear Stress	22217.32	1.000	24425.91	1.000
Tube Tensile Stress	12780.37	1.000	26850.38	1.000
Tube Compressive Stress	12780.37	1.000	39603.11	1.000
Tube-Tubesheet Joint load	12780.37	1.000	26850.38	1.000
Shell Stress (Axial, Junction)	3775.81	1.000	6069.84	1.000
Tube Pressure Stress	27453.63	1.000	19593.47	1.000
Tubesheet Extension Stress	2336.28	...	No Calc	No Calc
Minimum MAPnc	2336.28		6069.84	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0094	0.0094	...	...	...	...
2	-0.0663	-0.0664	...	...	...	...
3	-0.0569	-0.0570	...	...	...	...
4	...	...	...	...	...	...
5	0.0094	0.0094	-1.1668	-1.1683	...	...
6	-0.0663	-0.0664	-1.1668	-1.1683	...	...
7	-0.0569	-0.0570	-1.1668	-1.1683	...	...
8	...	...	-1.1668	-1.1683	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
= max(tubesheet thk/4, min(tubesheet thk, shell thickness))
= max(60.0/4, min(60.0, 9.53))
= 15.000 mm

Thickness Ratio = 0.856, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -16 °C
Min Metal Temp. at Required thickness (UCS 66.1) -24 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Channel Side MDMT calculation:

Governing thickness for the channel side:
= tubesheet thickness/4
= 60.0/4
= 15.000 mm

Thickness Ratio = 0.856, Temperature Reduction per Fig. UCS 66.1 = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -16 °C



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ASME TS Calc: DESIGN CASE

Case: 1 9:53a Aug 12, 2026

Min Metal Temp. at Required thickness (UCS 66.1) -24 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:

= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1

= max(2000.0/2336.28, 2600.0/6069.84)

= 0.856

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: